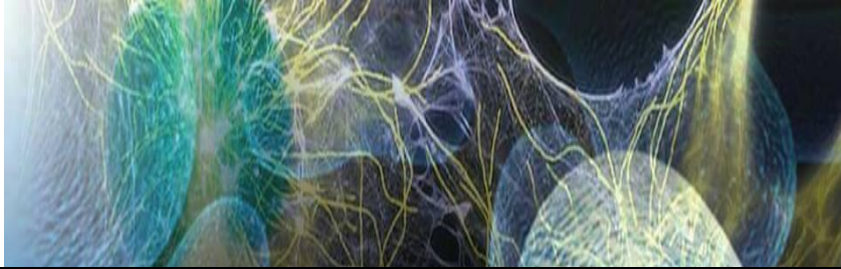


Healing Group

وعزمنّا أن نحسن سعيّا



Micro Lab (Final) 1-2

2017

Microbiology Lab



Done by: Mohamad BshaiRi

كلية التمريض



Purification and identification of bacteria



Purification of bacteria

* contaminated culture (less than **3** colonies)



- culture that contains bacteria from different sources.

* We have different shapes of bacteria, that means we have different types of them, then we make purification to develop a pure culture

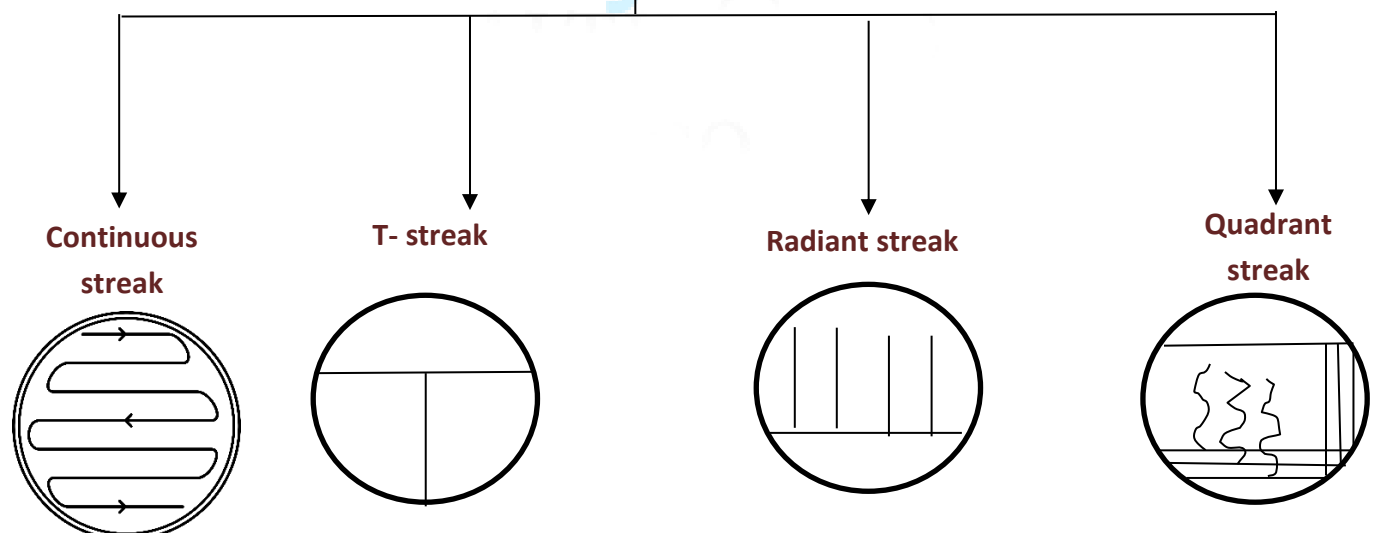


* Non- contaminated:
culture contain from same sources

(Culture that contains
one type of colony,
one type of bacteria)

* We have four methods for purification:

Purification method



* Quadrant streak method :

Name two tools you used to develop a pure culture through Quadrant method ?

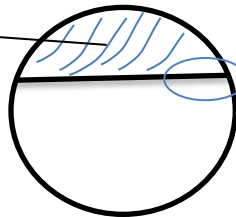
1. Needle
2. Loop

1. By sterile cool needle, put small amount of colony and draw straight line on agar edge

In colony agar

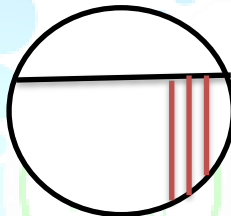
Cooling area

أقل عدد بكتيريا

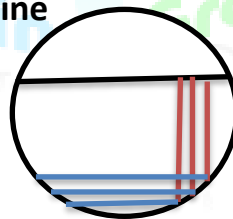


In cool area

2. By sterile cold loop draw **3** straight lines from first line



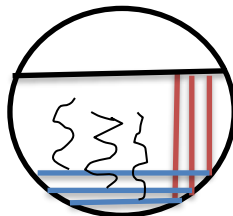
3. Resteralize the loop and check it cool in cooling area, then draw **3** straight lines from the corner of previous line



Above 45°c agar dissolve

حتى أقل عدد بكتيريا عليه

4. Resteralize the loop and check it cooling, then draw **3** zig zags



(To increase surface area of streaking)

5. Place the plates into the incubator (upside down) ← بنحط الـ agar

*identification of bacteria :

1. Microscopic characters for bacteria (gram stain for each plate)

*Shape of bacteria



Cocci



bacilli



spirilli



vibrio

* arrangement of bacteria

- if we have one cell it's single arrangement →

- if we have two cell it's diplococci →

- if we have 4 cell it's tetrad →

- if we have 8 cell it's sarcinae →

- if we have more than 8 cell it's staph (grape like) →

- if we have chains (streptococci) →

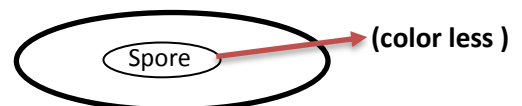
- if we have  **Lactobacilli**

* Group of bacteria

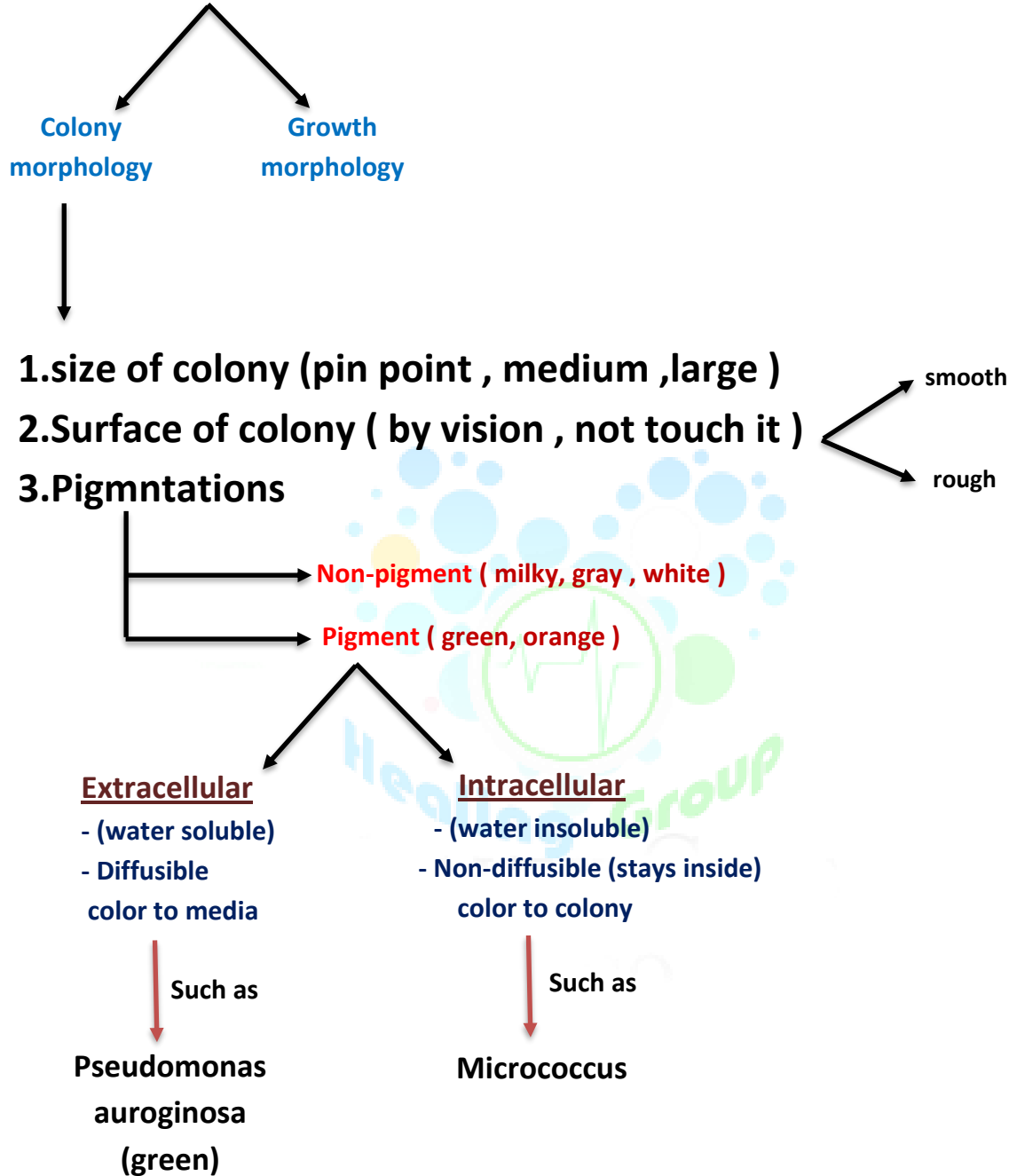


* spore form or not spore form

* position of spore



2. Macroscopic characters for colony:



4. Form: we determine

- Configuration
- elevation
- edge of colony (Margins)

موجودين بصفحة ١١

Sheet

Each group must have at least one non-contaminated culture of isolated colonies through serial dilution with surface spreading methods or pour plate methods then for each culture do the following:

* **Study the macroscopic characters for the bacteria which includes the followings:**

a- Colony morphology

b- Growth morphology

* **COLONY MORPHOLOGY:**

- Study the colony morphology according to the following parameters:

1* **Size of the colony:** the colony may be pinpoint, small, medium, or large

2* **The surface of the colony:** by vision and not by touching.

- look to the surface of the colony, if the surface contains warts and/or beads, it's rough, otherwise the surface of the colony is smooth

* **Pigmentation:** the bacteria may be Pigment producer or not.

* Pigment producer:

- The bacteria may be produced by the bacterial cells and seized inside the cells which became colored and result in colored colonies, this pigment is in diffusible water insoluble pigment. So, this pigment is a **color of the bacterial colonies**.

Micrococcus produced on nutrient agar:

Yellow in diffusible water, Black in diffusible Water, orange in diffusible water, or...etc.

- If the bacteria synthesis the pigment and secret it outside to change **the color of the media** around the colony, it is called **diffusible water-soluble** pigment such as **Pseudomonas aurogenosa** which secrete green diffusible water-soluble pigment on nutrient agar or broth called pyocyanin.

- The bacteria may be Produced one of the above pigments or the two ,To gather also the pigment production, color and the type depends on the media where the bacteria cultured.

* Forms: According to the followings:

- 1- **Configuration**
- 2- **Edge of the colony**
- 3- **Elevation**

As it is clear in the drawing in attached paper.

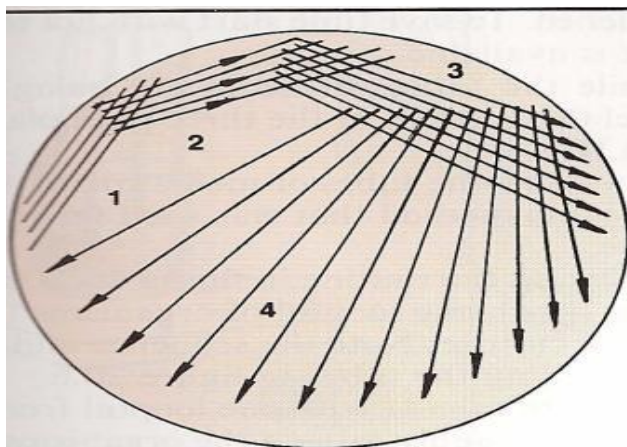
The colonies for each type of bacteria must be identical not similar.
After that you must divide the colonies on the non-contaminated culture of isolated bacterial culture into groups and the colonies of each group must be identical.

**** Take one colony for each group and start to purify it in one of the following methods of surface streaking methods.**

*** SURFACE STREAKING METHODS :**

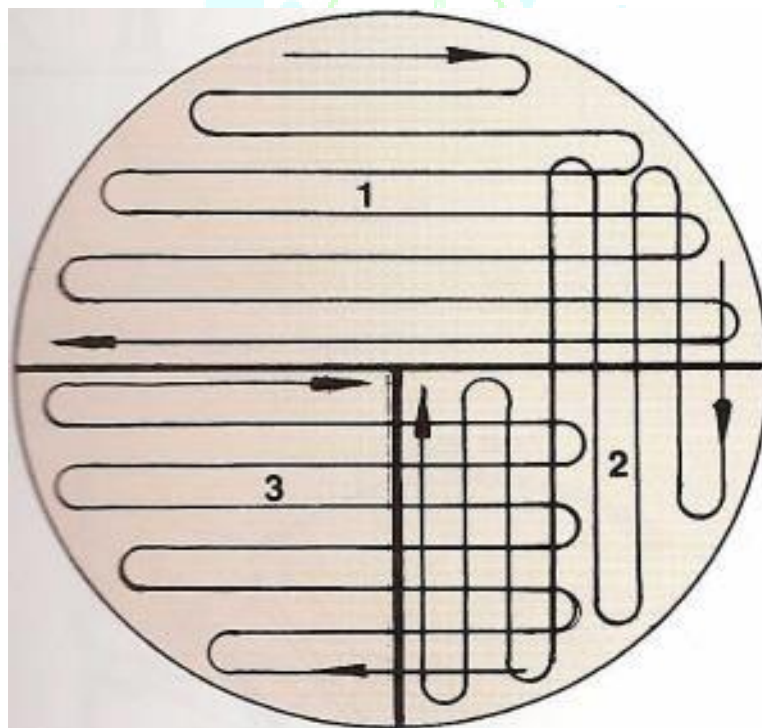
A) QUADRANT STREAK

- 1- Streak one **loopful** of organisms over Area **1** near edge of the plate.
Apply the loop lightly.
Don't gouge into the medium.
- 2- Flame the loop, cool **5** seconds, and make **5** or **6** streaks from Area **1** through Area **2**.
- 3- Flame the loop again, cool it, and make **6** or **7** streaks from area **2** through Area **3**.
- 4- Flame the loop again and make as many streaks as possible from Area **3** into Area **4**, using up the remainder of the plate surface.
- 5- Flame the loop before putting it aside.



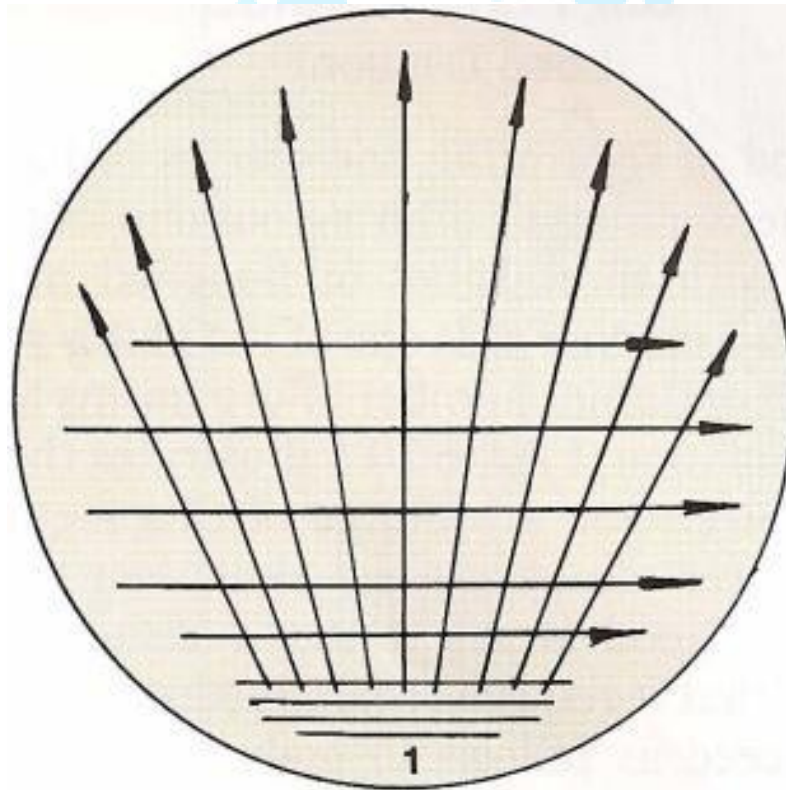
B) "T" STREAK

- 1- With marking pencil draw a "T" on bottom of plate dividing the plate into one half and two quarters.
- 2- Inoculate the half portion with one loopful of organisms in a continuous line from **edge of the plate to midline** of the plate.
Don't gouge the medium.
- 3- **After flaming the loop**, cross-streak from Area 1 into area 2 with one continuous streak, filling Area 2.
- 4- **After flaming loop again**, cross-streak from Area 2 to area 3 with a **single** continuous streak.



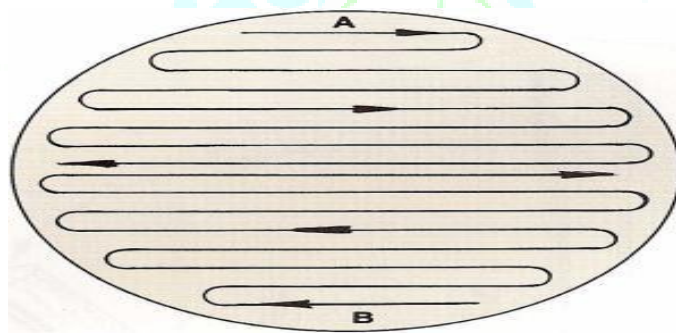
C) RADIANT STREAK

1. Spread a loopful of organisms in **small area** near edge 01 plate (**Area 1**).
2. Flame the loop and allow it to cool for **5** seconds.
3. From the edge of **Area 1** make seven or eight straight streaks **to the opposite side of the plate**.
4. Flame the loop again and cross-streak over the last streaks, starting near **Area 1**.
5. Flame the loop again **before putting it aside**.



D) CONTINUOUS STREAK

1. Starting at the edge of the plate (**Area A**) with a loopful of organisms, spread the organisms in a **single** continuous movement to the center of the plate.
2. Rotate the plate **180** degrees so that the uninoculated portion of the plate is away from you.
3. **Without** flaming loop, and using the same face of the loop, continue **streaking** the other half of the plate by starting at **Area B** and working toward **the center**.
4. Flame your Loop before setting it aside.



Then incubate it at optimal temperature over night.

After that look for **identical isolated** colony, if it's that the culture is **pure**,

Otherwise (no isolated colonies or/and the colonies is not identical, even if the culture contains all of the colonies identical except one pinpoint colonies) the culture **is not** pure and you must repeat the process till you get a culture of identical isolated colonies.

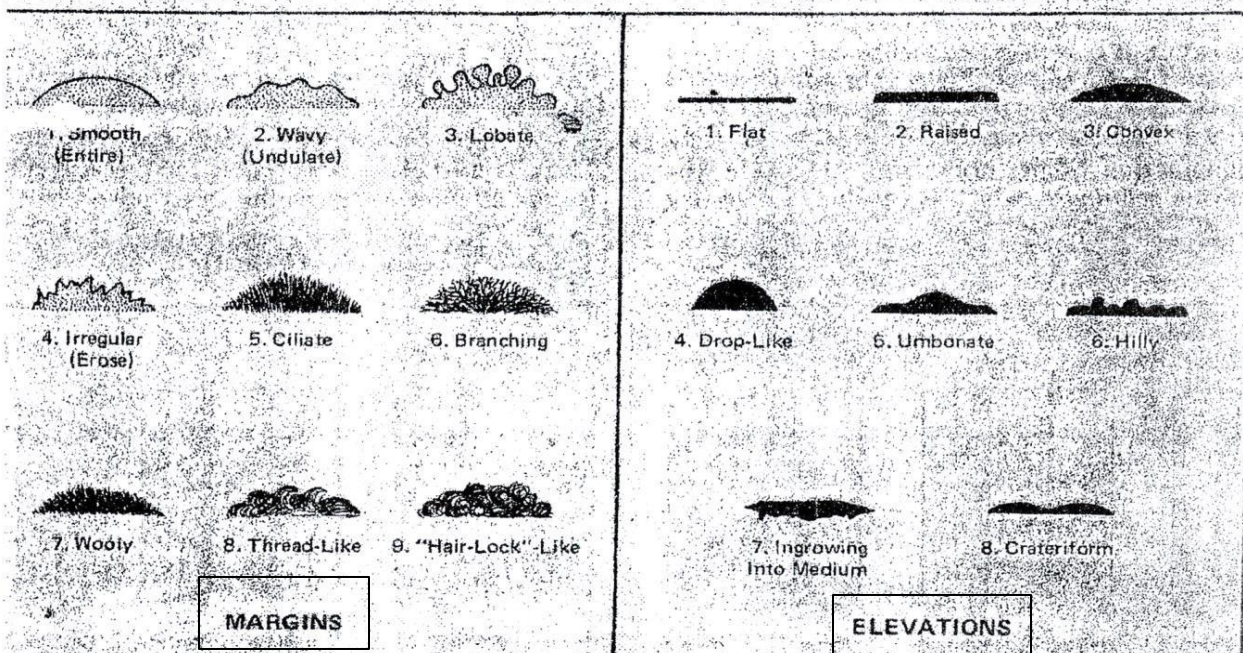
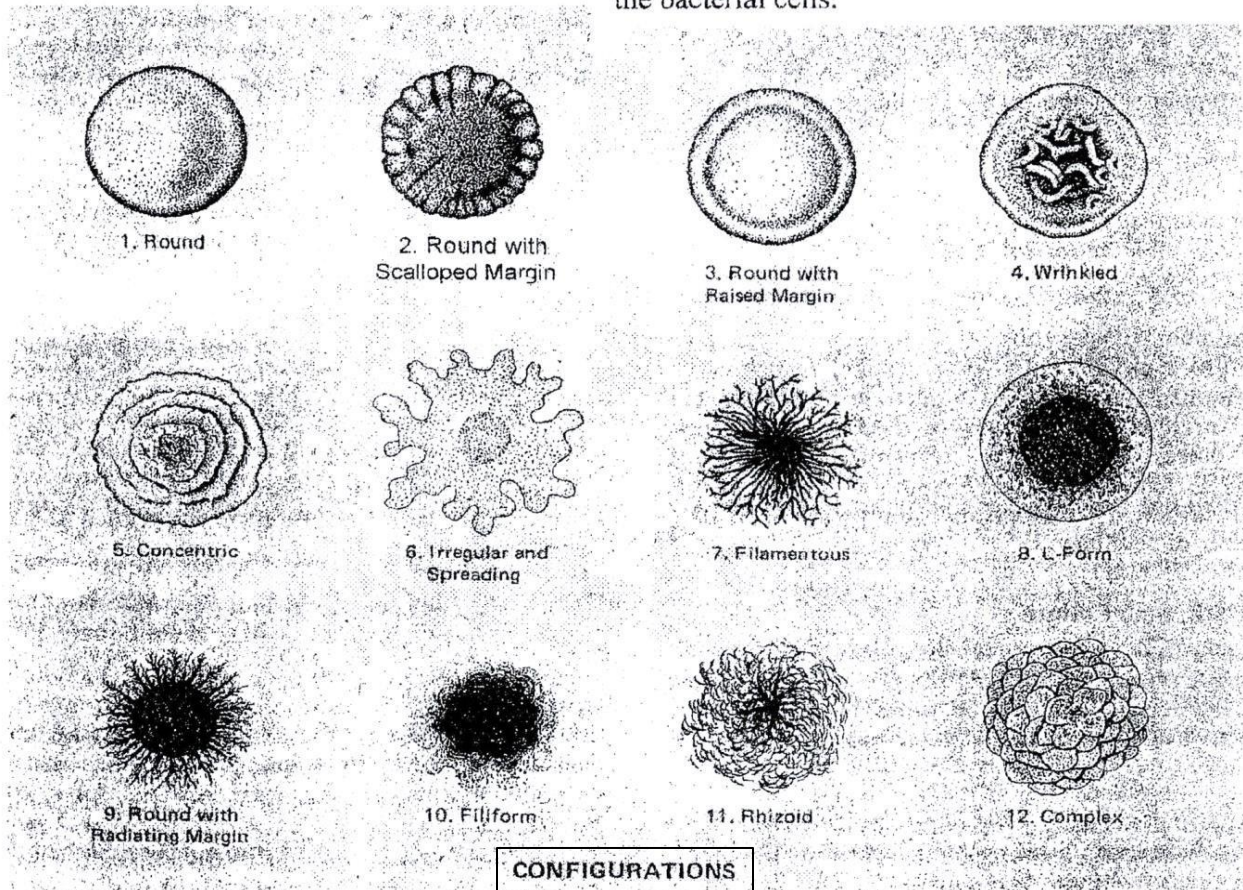
After you get a pure culture you must study the microscopic morphology:

Gram staining, acid – fast staining if the culture is needed, spore staining, negative staining... Etc.

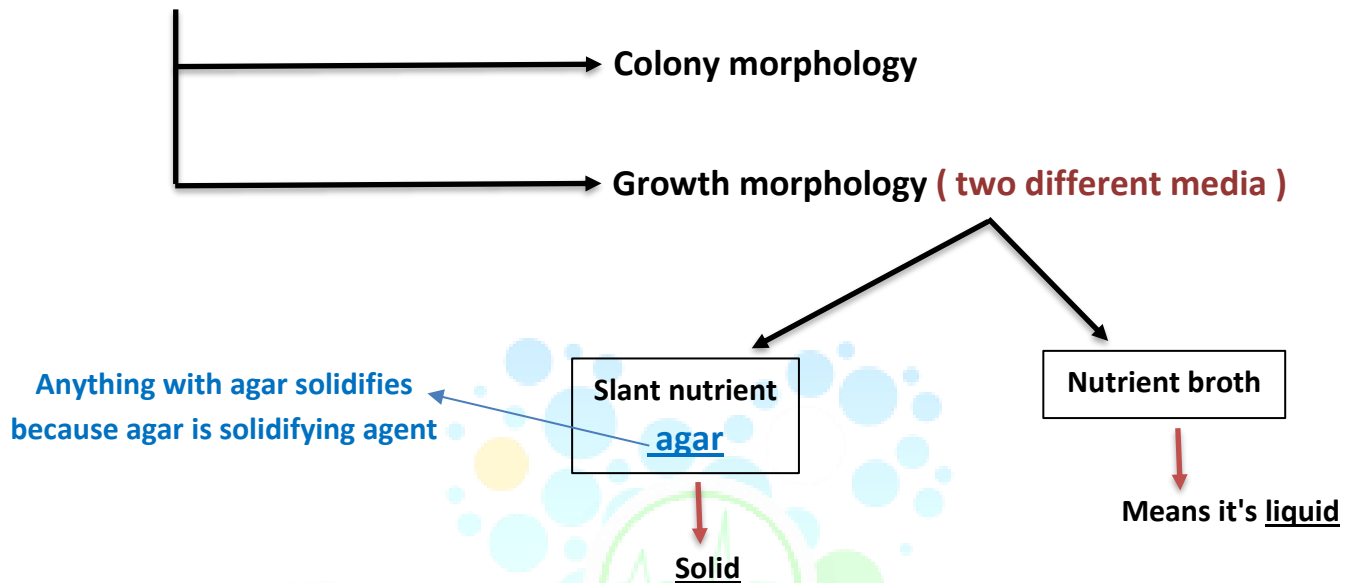
And you must determine the shape, size, and arrangement of the bacterial cells.

Forms

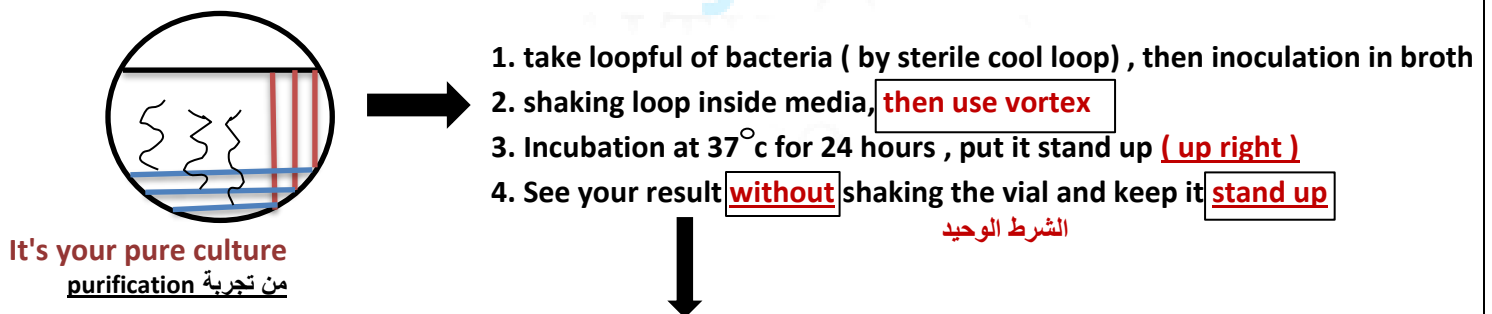
staining, negative staining etc. and you must determine the shape, size, and arrangement of the bacterial cells.



* Macroscopic characters:

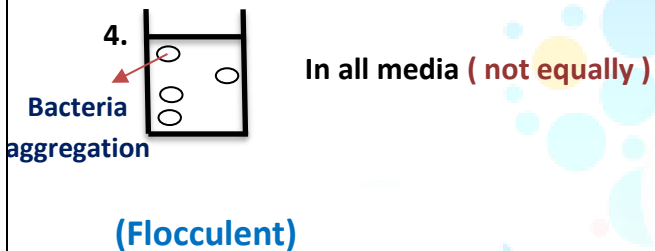
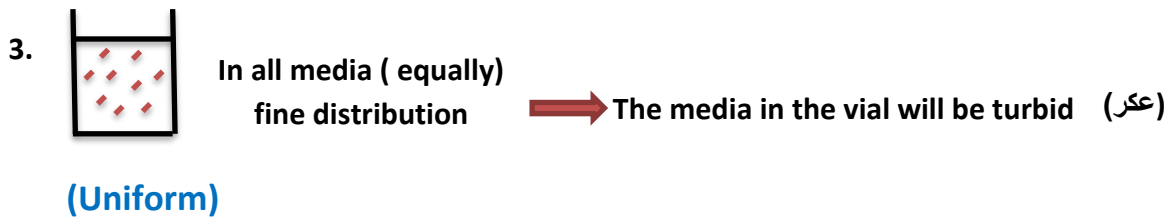
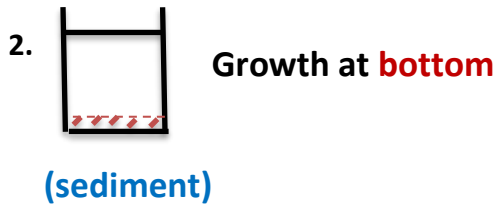


* Growth morphology on nutrient broth:

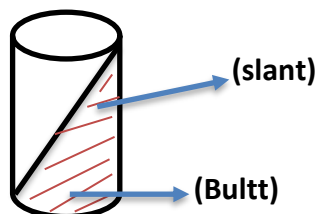


* The results according where the bacteria grew as following :

1. Growth in **surface**
(pellicle)



* Growth morphology on slant nutrient agar :



(لمرة واحدة فقط)
Single line inculation by loopful of bacteria from
your (pure culture) by sterile cooled loop
then incubate at 37°C for 24 hours .

Results according to the following parameters:

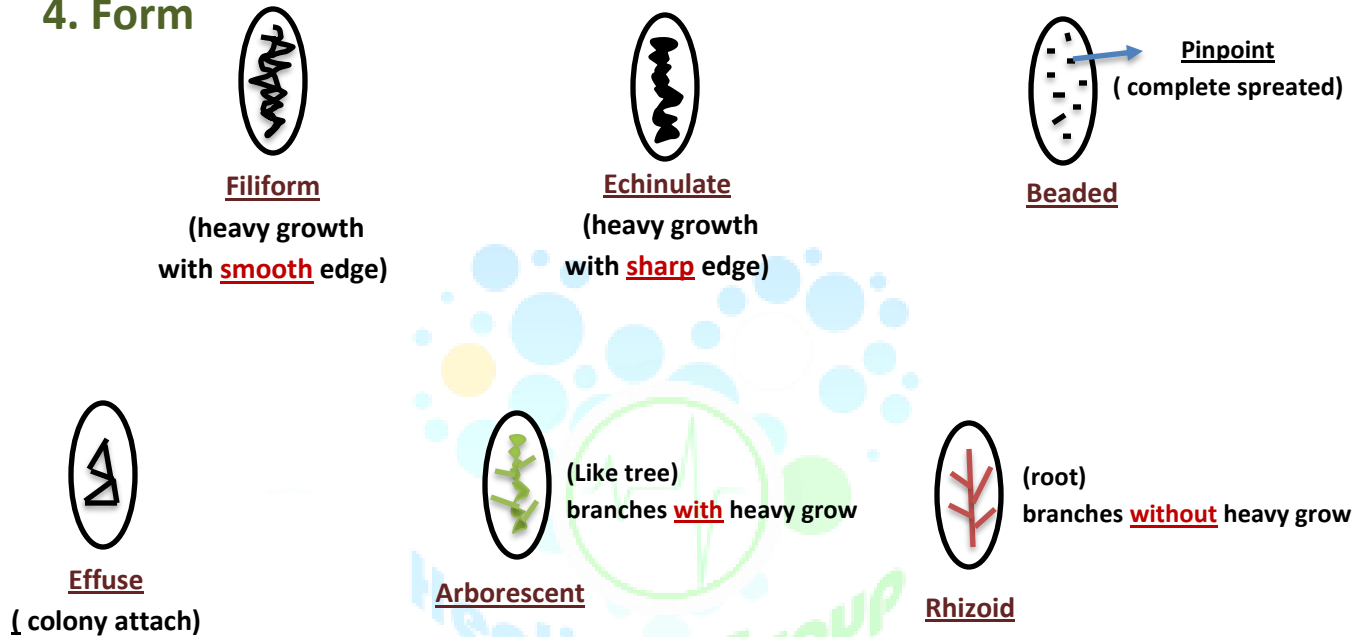
1. Abundance of growth (None, small, moderate ,large)

2. Pigmentation

3. Optical character

- **transparent** if it transmits light **100%** → (if we have **colorless** colony)
- **translucent** less than 100%
- **Opaque** it transmits **0%** (**black**)

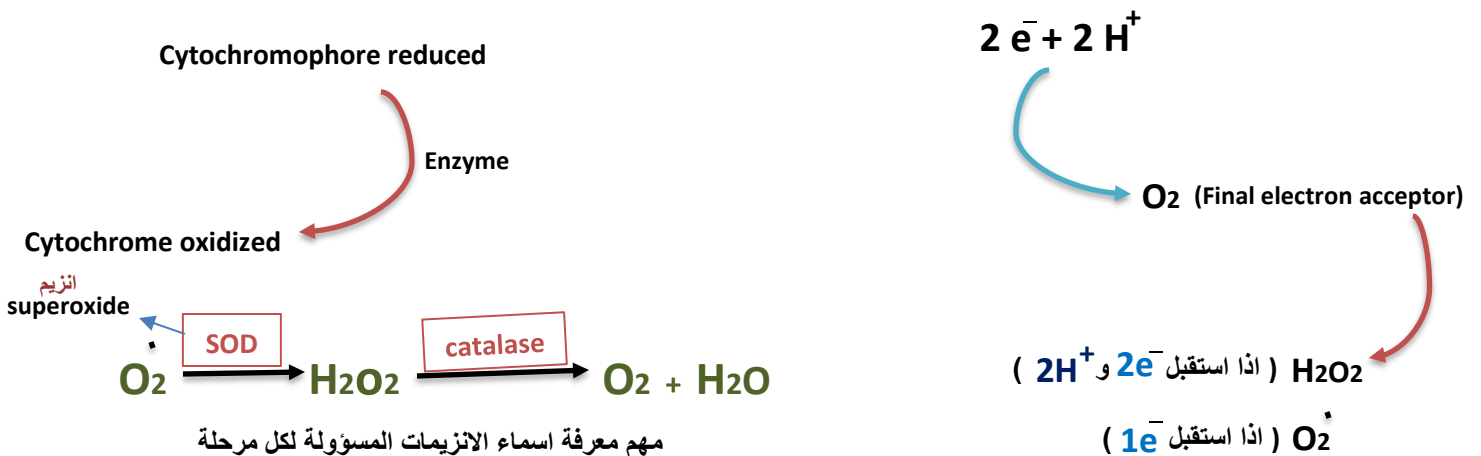
4. Form



* physical factors that affect bacteria growth :

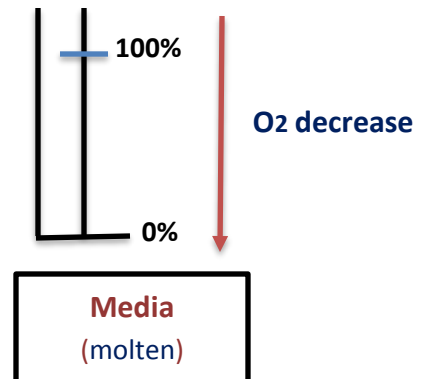
O₂ requirement (according to the present or absence of 2 enzymes)

* In electron transport chain



- we use brain-heart infusion agar (complex media)

because this media contains reducing agent of O₂



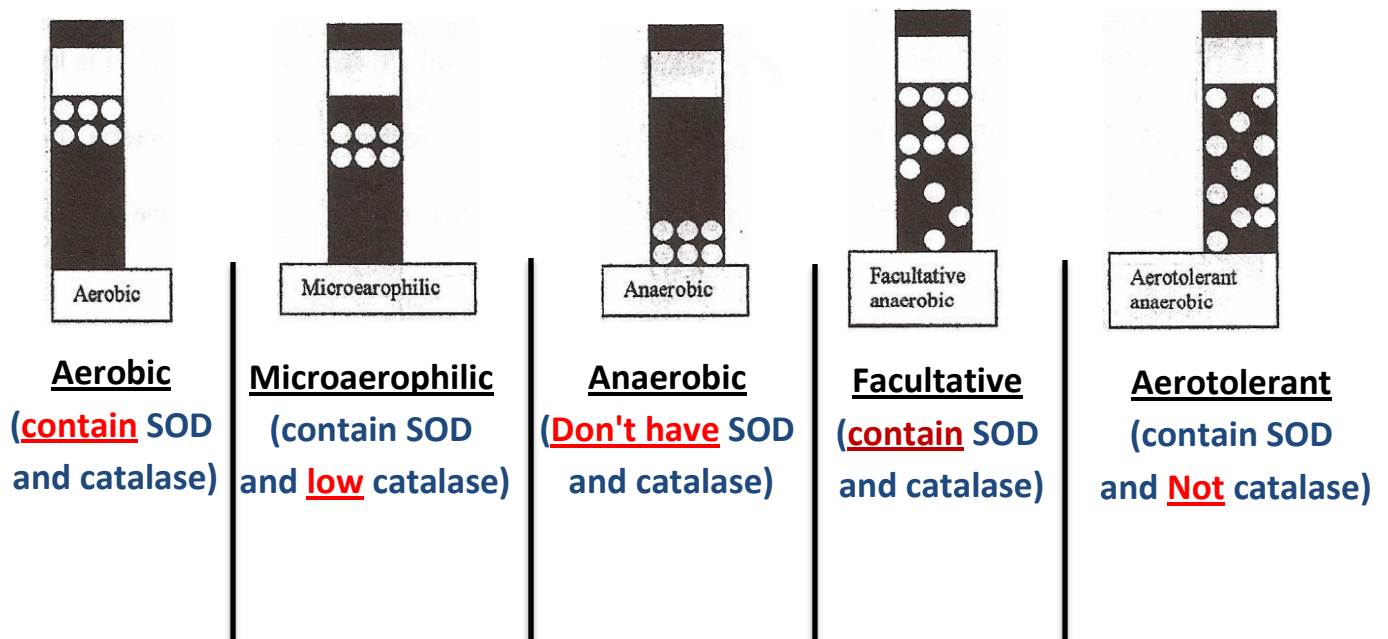
brain-heart infusion agar → inoculate a large loopful of bacteria

(cool to 45°C)
(holding temp)

Vortex (uniform distribution)

- * Immediately put it under tap to cool it
- * incubation for 4 days .

*Results :



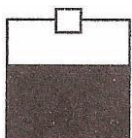
Sheet

1. Macroscopic characteristics includes:

- Colony morphology
- Growth morphology

The study of bacterial growth on Nutrient broth and Nutrient agar slant:

Nutrient Broth:



Inoculate a vial of nutrient broth with a loopful of bacteria from your isolate then shake it by vortex very well to get uniform distribution of bacteria and incubate it up right for 24 hours then take the results according where the bacteria grew as following:

2. On Nutrient area slant:



Pellicle
On the surface



Sediment
In the bottom of the media



Uniform fine distribution
Throughout the media



flocculent uniform with bacteria cell aggregation



Inoculate the **surface of the nutrient agar slant** by sterile cooled loop from your **pure culture** through **single line** inoculation then incubate it for 24 hours then take the **results** according to the following parameters:

3. **Abundance of the growth:** Non, slight, moderate, large

4. **Pigmentation:** diffusible and in diffusible pigments

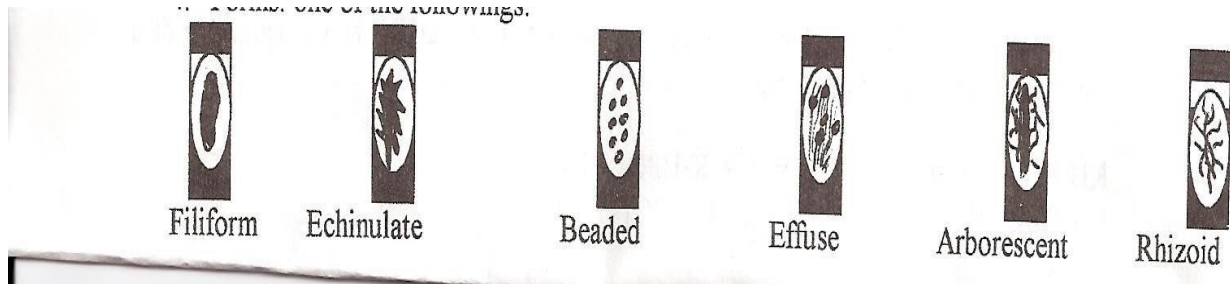
5. **optical characters:** the amount of light that can be transmitted through the bacterial growth, so the bacteria may be:

Transparent: the light pass through it (100 %)

Translucent: the light pass through the bacterial growth partially (less than 100 %)

Opaque: No light transmit through the bacterial growth

4. Forms: one of the followings:



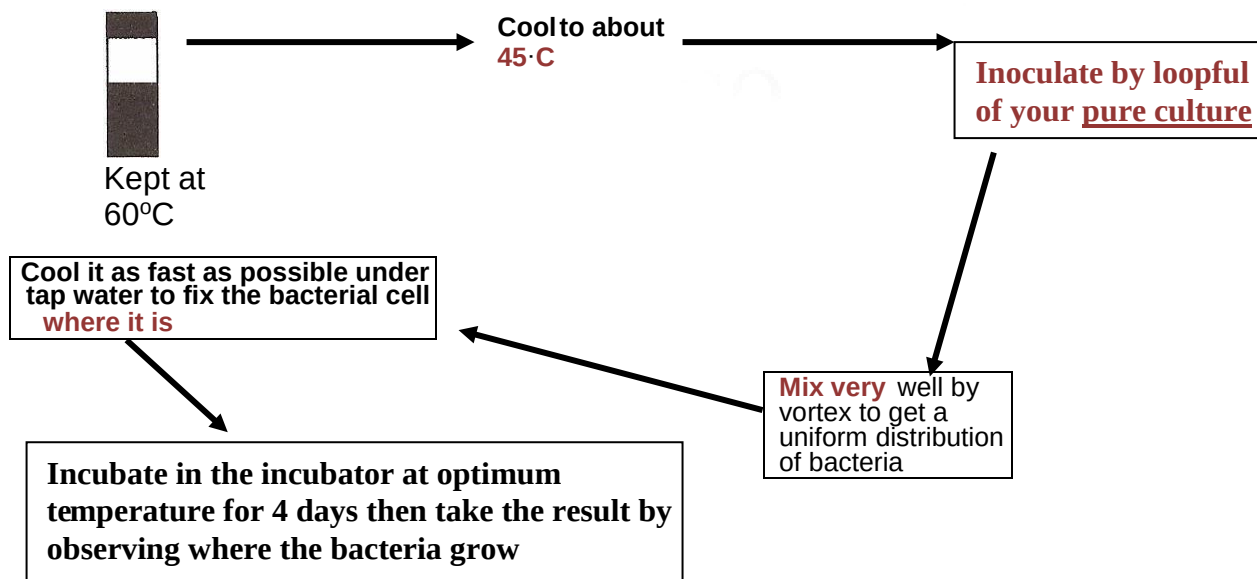
Physical factors:

There are many factors affect the growth of bacteria by affecting one or more than one of the followings:

1. Cell wall: inhibit the synthesis of cell wall or breakdown it.
2. Cell membrane: by affect the integrity of it.
3. Cell cytoplasm by affect the colloidal structure of cytoplasmic proteins
4. Genetic material
5. Enzymes.

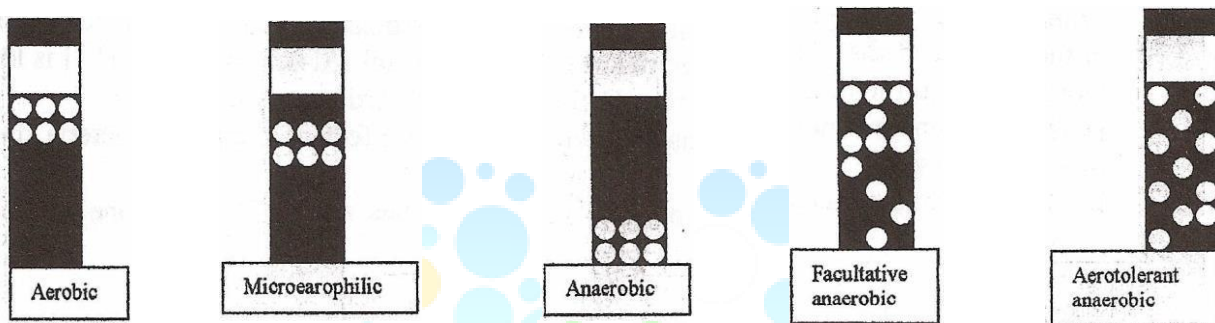
O2 requirement:

You have a molten Brain - heart infusion agar in test tube at 60 °C



Results as one of the following ...because

the bacterial cell will grow to colony if the amount of oxygen is suitable



please forgive us for any mistake!

Nursing group

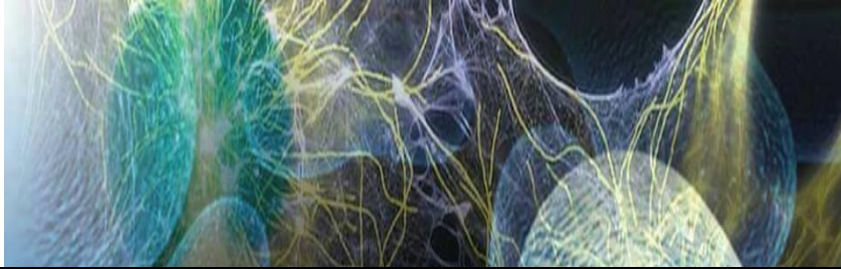
Healing group



تم بحمد الله

Healing Group

وعزمننا أن نحسن سعيًا



Antibiotic (Final)

2017

Microbiology Lab



Done by: Mohamad BshaiRi

كلية التمريض

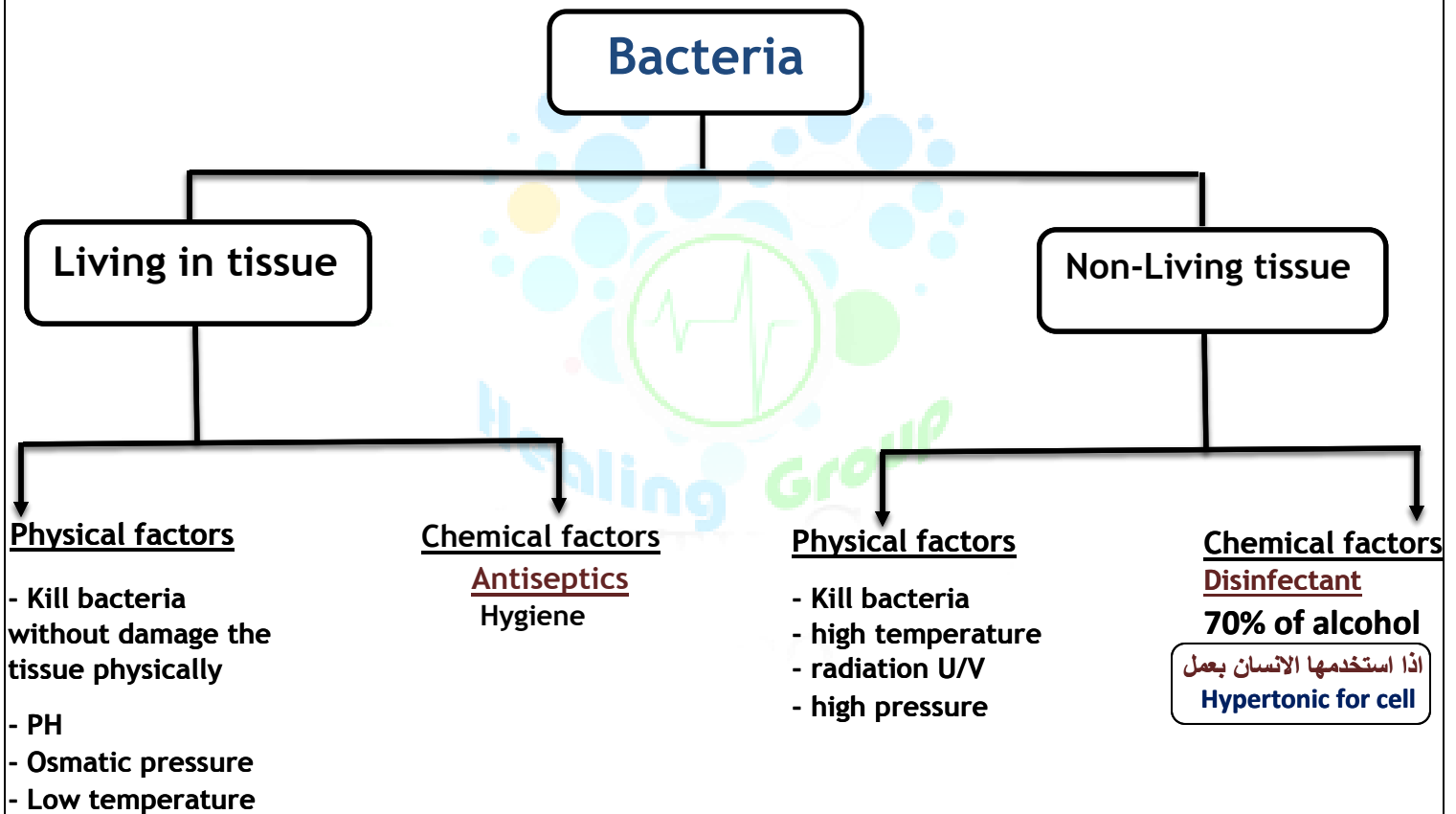


Antibiotic



* Factors affect bacterial growth :

1. Physical factors
2. Chemical factors



* We divide the chemicals according to their actions into two types:

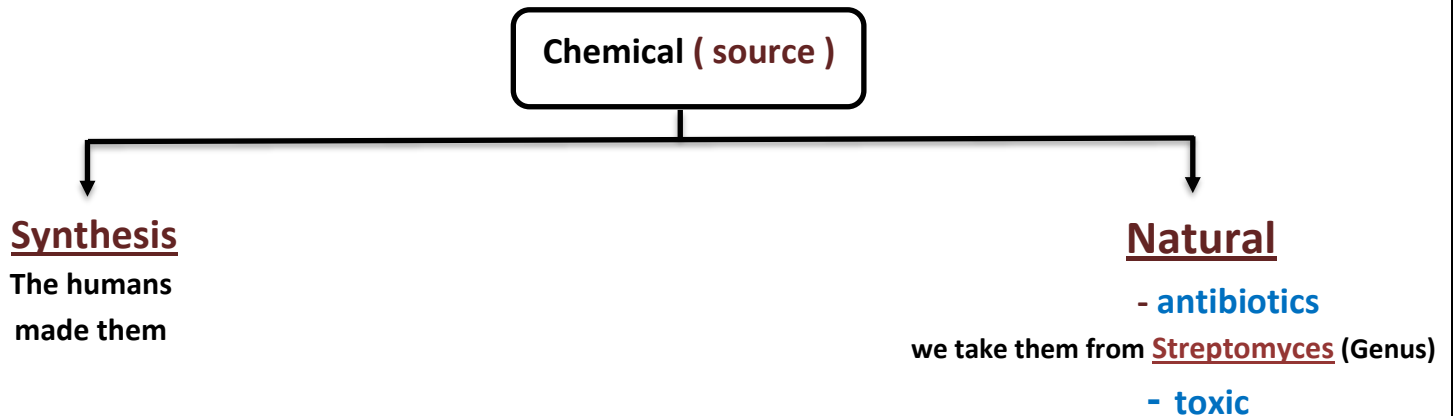
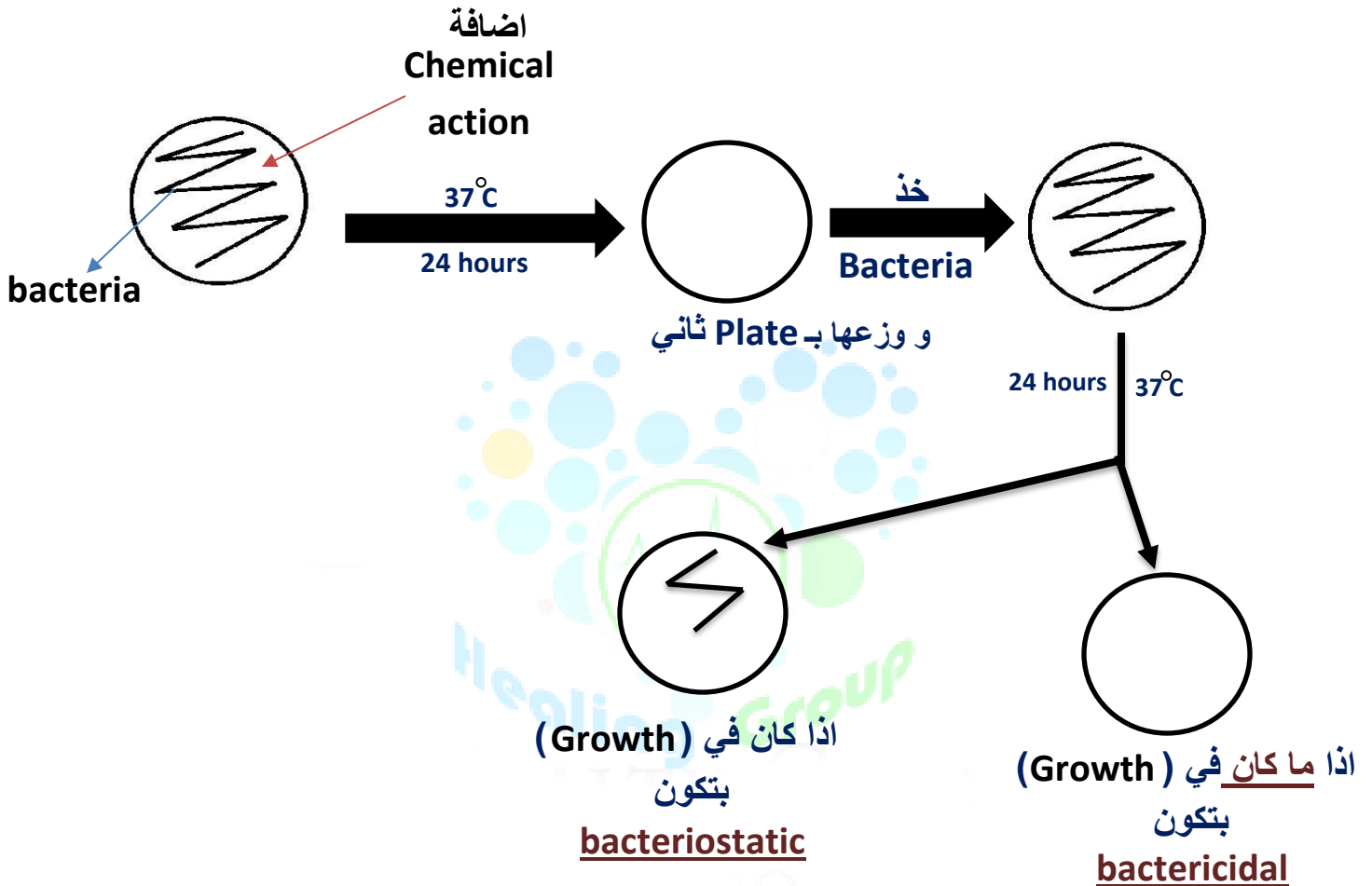
1- **Bactericidal** → Stop bacteria growth with kill it
(kills the bacteria)

2- **Bacteriostatic** → Stop bacteria growth without killing it
(inhibit bacterial growth)

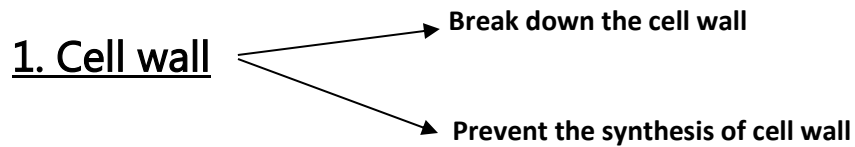
when this factor is remove, it retain

Cidal → Kill
Static → inhibit

* We know this chemical is bactericidal or bacteriostatic by following the steps :



* Site where the antibiotic affected the bacteria (How antibiotic affect bacteria) :



2. Plasma membrane

- Breakdown plasma membrane
- prevent the synthesis of plasma membrane material
- permeability (increase , decrease)

3. Protein synthesis (enzyme)

- Ribosomes (inhibit synthesis)
- mRNA (transcription)
- tRNA (translation)

4. DNA replication

5. Metabolic pathway (glycolysis) → ATP

* Disc diffusion method → (Kirby-Bauer method)

* Principle we should follow it : (factors affect test)

- 1- **Type of antibiotic**
- Broad spectrum (Affect high numbers of bacteria)
 - Narrow spectrum (Only one site)

2- Type of the media (Mueller Hinton agar)

- we use this media → because it is pure media and give high diffusion rate of antibiotics .

3- Type of bacteria

4- Titers of bacteria

→ Concentration
(↑ titer → effect of antibiotic ↓)

Optimum titer (10^7)

كل ما زاد تركيز البكتيريا يقل تأثير الانتي بيوتك

5- If the rate of diffusion higher than the rate of bacterial growth the accuracy of our result more than 1 → **Correct**

يعني شغلك صح

≥ 1

6- concentration of antibiotic

If ↑ → toxic

If ↓ → Not effect

7- Rate of diffusion antibiotic

8- Growth Rate of bacteria

$$* \frac{\text{Rate of bacterial growth}}{\text{Rate of antibiotic diffusion}} < 1$$

$$* \frac{\text{Rate of antibiotic diffusion}}{\text{Rate of bacterial growth}} \geq 1$$

* Antibiotic sensitivity test :

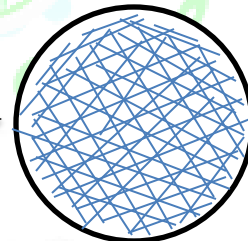
- Main purpose → to determine strongest antibiotics that affect bacteria growth



Inoculate by
loopful of bacteria

Vortex

Mueller Hinton agar media

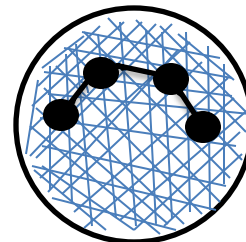


بنوخذ مرة او مرتين
اقصى حد من البكتيريا
ونزعههم ع اكبر
مساحة من الميديا

Heavy inoculation
using cotton swap

Add antibiotic disk
using forceps

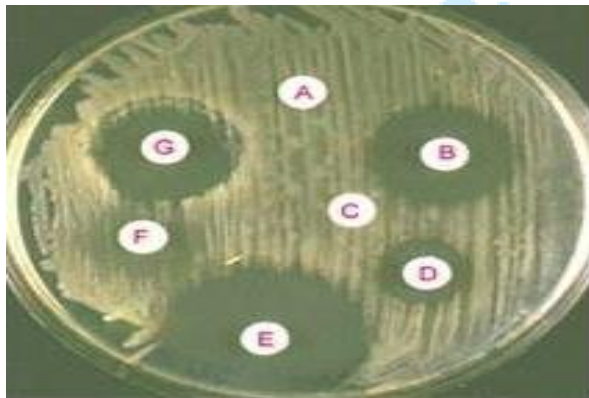
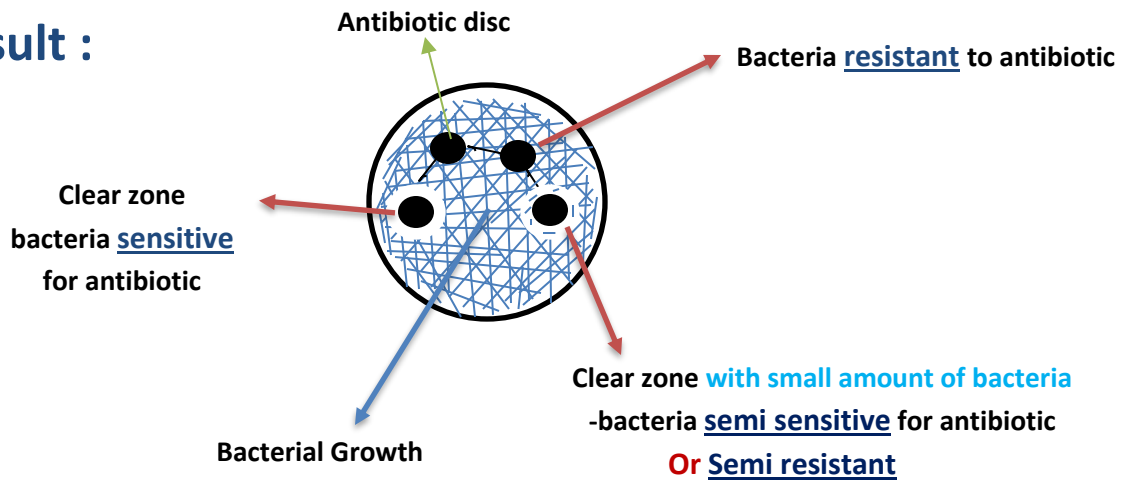
عدهم اربعة كل وحدة
تختلف عن الثانية



Result

37°C
Incubate over night
24 hours

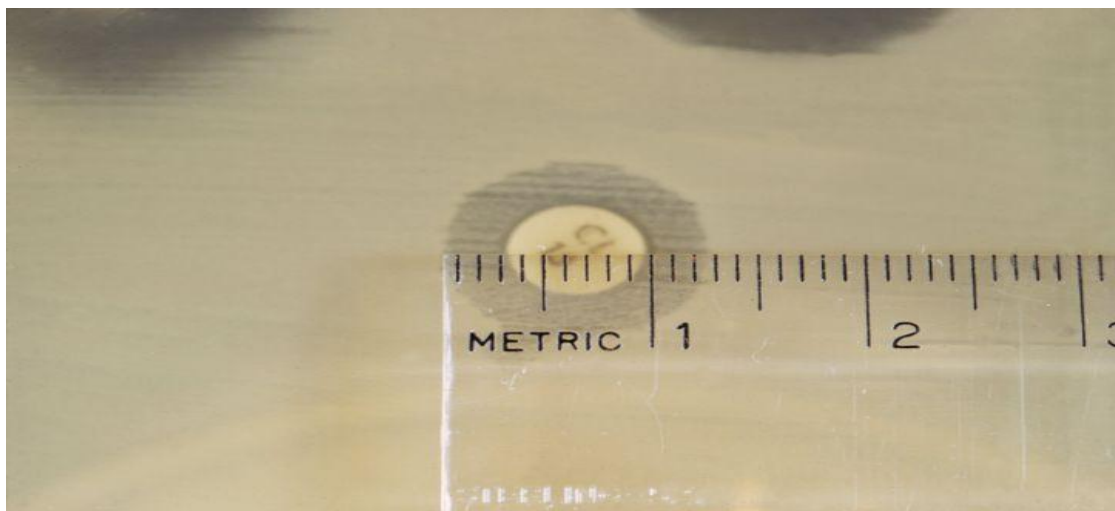
* Result :



If we have clear zone around bacteria mean there is killing

Bacteria sensitive for antibiotic

* we use **rule** to measure **zone**



مثال

	Measure zone	Standard zone	Real zone
A	0.0	1	-1
B	2 mm	3	-1
C	6 mm	5	1
D	3 mm	1	2

* Real zone = measure zone – standard zone

من خلالها نستطيع تحديد فعالية المضاد الحيوي

* Measure zone (by role) → ما بنقدر نحدد فعالية المضاد الحيوي من خلاله

please forgive us for any mistake!

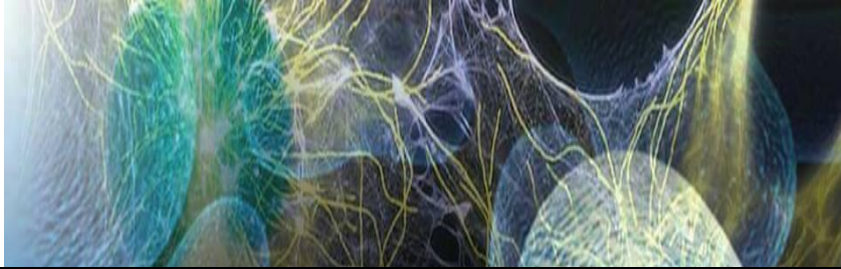
Healing group " Nursing group "



تم بحمد الله

Healing Group

وعزمنّا أن نحسن سعيّا



Media (Final)

2017

Microbiology Lab



Done by: Mohamad BshaiRi

كلية التمريض



Media



Media :

Nutrients, carbon source, vitamins , ions , salt , water

Sugar, protons, bacteria use ATP synthesis

* Media according to its composition:

1- Defined media (simple):

we know the exact concentration of each component.

Ex: Nutrient agar , Nutrient growth

2- Complex media: we don't know the exact concentration of each component.

(can hardly synthesis in the lab)

Ex: - brain – heart infusion agar.

- S-S media
- Bismuth agar
- TSI agar

If media name contain any of these words it is complex Media

(Beef , tissue , infusion , meat , yeast , extra , soay)

*Media according to shape:

1- plate



2- slant (solid)



4- vial (Liquid)



Agar



Is solidifying agent

*Media according to status:

- 1- **Solid** → agar ^{More} $\geq$ 1% (brain – heart , nutrient agar)
- 2- **Semi-Solid** → agar less than 1% (SIM → Sulfide indole motility)
- 3- **Liquid** → agar is 0% (broth)

*Media according to purpose:

- 1- **General media** → that allow all bacteria to growth

Ex (Nutrient agar , Nutrient broth)

- 2- **Selective media** → that allow some bacteria to growth and inhibit other type of bacteria.

Ex: (F-agar) → selective for Pseudomonas

(coli form) → selective for E. coli

(chocolate) → selective for Neisseria

- 3- **Enrichment media** → that all bacteria growth but one more other...

ex : (lactose growth media for salmonella)

- 4- **Differential media** → that differentiates between bacteria according to color,...

- 5- **Selective – differential** ex: (MacConkey, Hekton , EMB , XLD , TSI , MSA, Bismuth...)

S-S media → selective for salmonella and Shigella

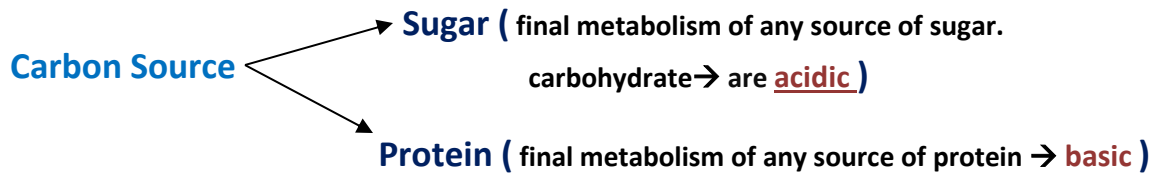
- 6- **Enrichment – differential media** → (blood agar)

- 7- **General - differential media** → Nutrient agar

Pseudomonas → Green

*Selective-Differential Media: (6 media)

1- MacConkey's agar:



*** All media are prepared on neutral PH ***

*** Component of MacConkey:**

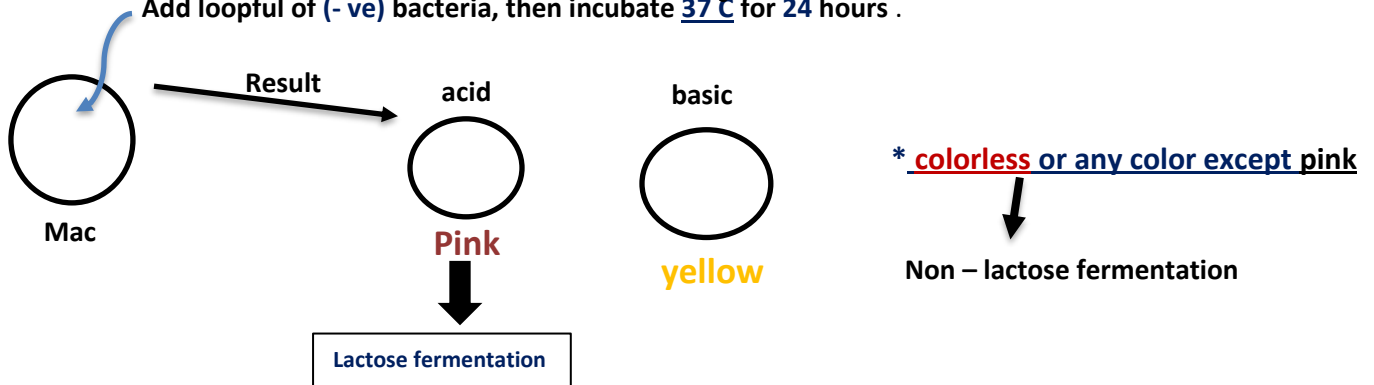
- Crystal violet → Inhibits growth of gram positive
- Bile salt → Inhibits growth of gram positive except two types: 1. Enterococcus
2. Staphylococcus
- Lactose (sugar) and pepton (protein) → (C-source)
- PH-indicator (Neutral red)

***Neutral ... red**

***Acid ... pink** → sugar بنعرف انه بكسر

*** Basic ... yellow** → protein بنعرف انه بكسر

Add loopful of (- ve) bacteria, then incubate 37 °C for **24** hours .



If color is pink with glisten (لامع) → it mean Klebsiella bacteria

هي النوع الوحيد الي بنقدر نحدده

2- Eosin methylene blue: (EMB)

- Components:

* Eosin and methylene blue, Two function:

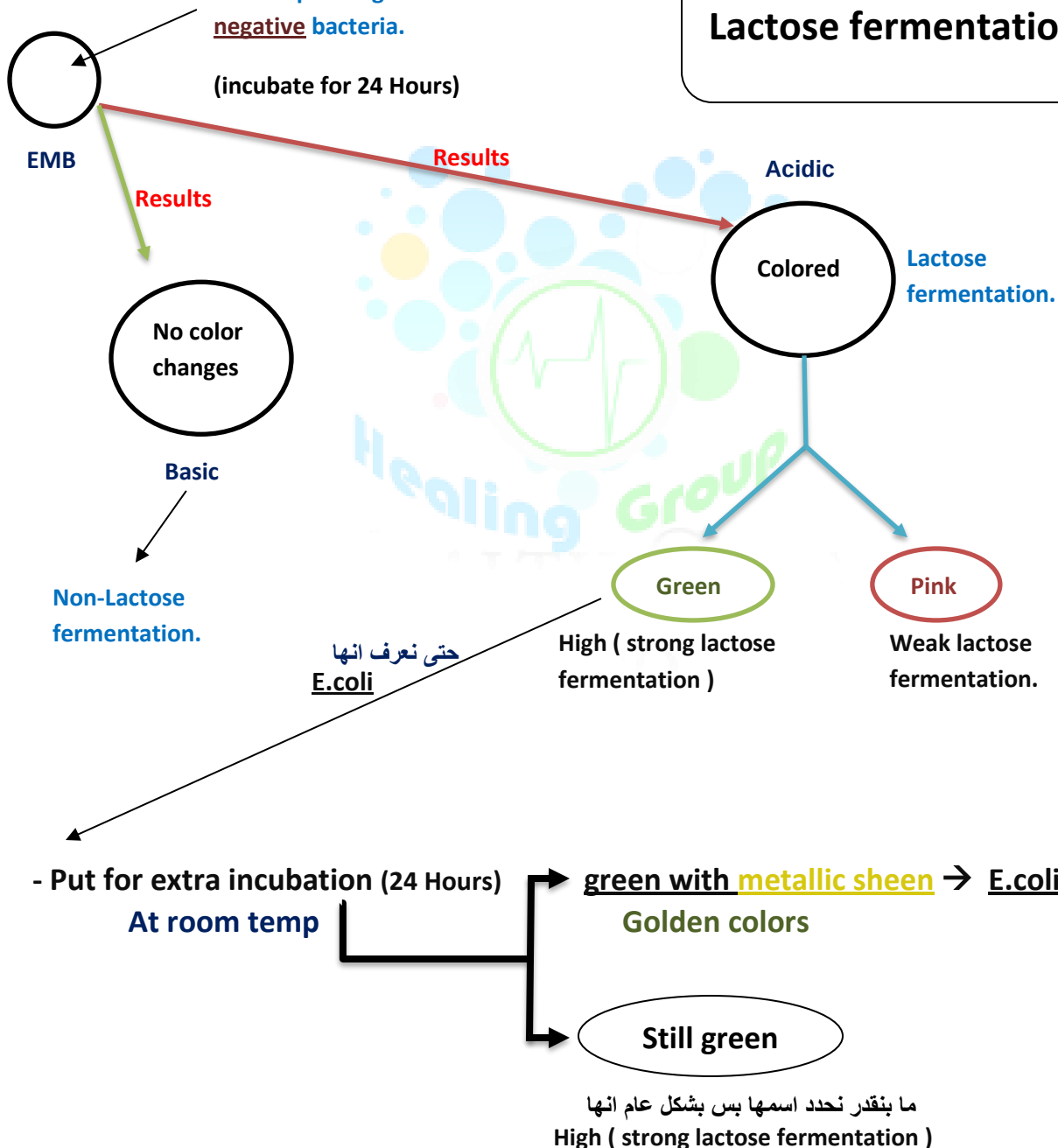
1. Inhibit growth of gram-positive bacteria
2. React with acidic products to get color precipitation (any color)

* Lactose

Add loopful of gram
negative bacteria.

(incubate for 24 Hours)

E. coli is strong
Lactose fermentation



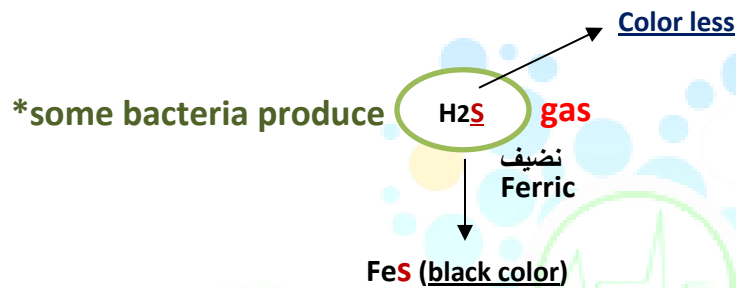
C-source → carbon source

3- xylose-lysine deoxycholate agar (XLD)

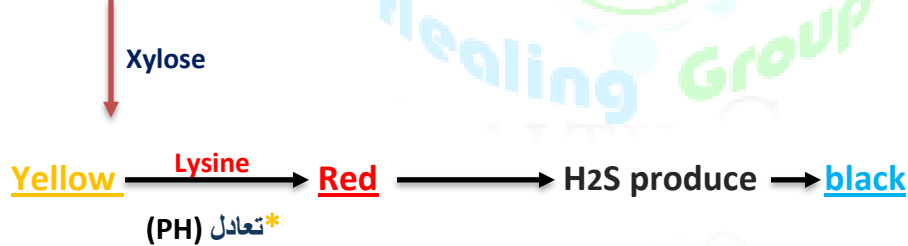
(salmonella and shigella)
(coli forms)

- xylose, sucrose, lactose → Sugar (C-source)
- lysine → amino acid.
- Ph indicator (**phenol red**) → acid "**yellow**" / neutral "**red**" / basic "**pink**"
- **sodium thiosulfate** (sulfur source)
- **ferric ammonium citrate** (sulfur indicator)

انتبه
هون
اللون
عكس
Neutral
red



***salmonella** → xylose , lysine , Non Lactose , Non sucrose.



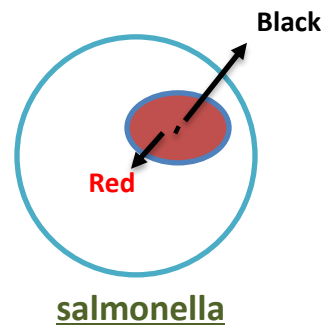
*تعادل (PH)
هو السبب في ظهور اللون الاحمر مش الزهري

***Shigella** → still red

Red

Smooth colonies

Non lysine , Non Xylose,
Non lactose , Non H₂S

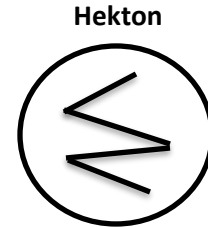


* coli forms (E.coli) → strong Lactose (yellow).

***Pseudomonas** → Lysine fermentation (Pink).
Rough colonies

4- Hekton agar :

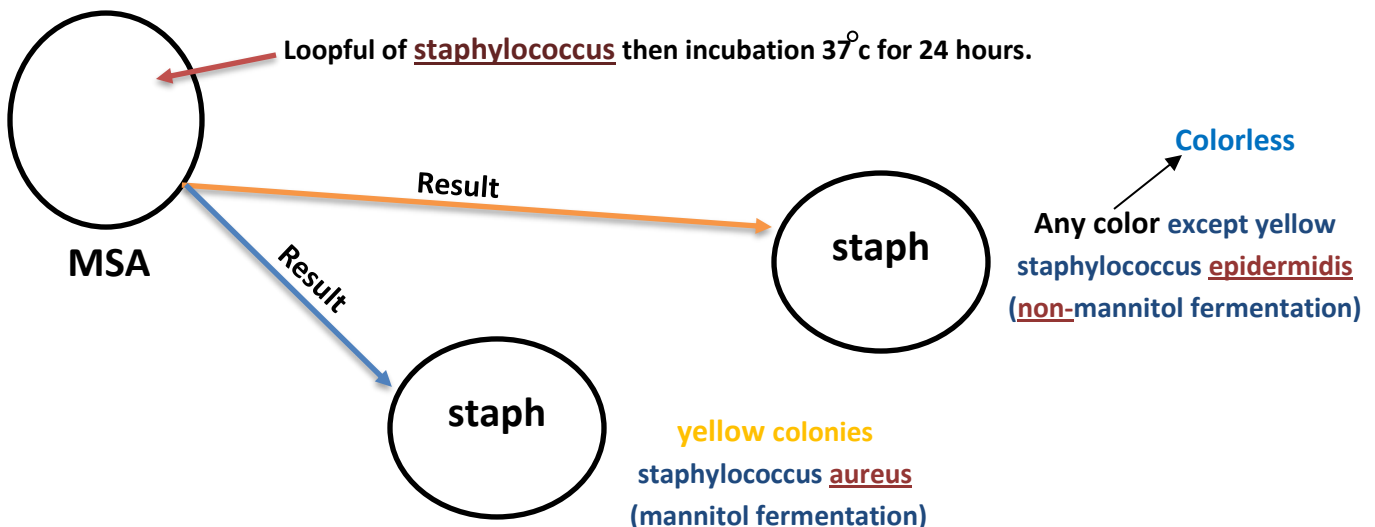
- sucrose, lactose, salicin → (C-source)
- Bile salt → inhibit growth of gram positive.
- PH inductor → Bromthymol blue → neutral → green
Acid → yellow
Basic → blue
- sulfur indicator → ferric ammonium citrate.
- sulfur source → sodium thiosulfate.



5- mannitol-salt agar (MSA)

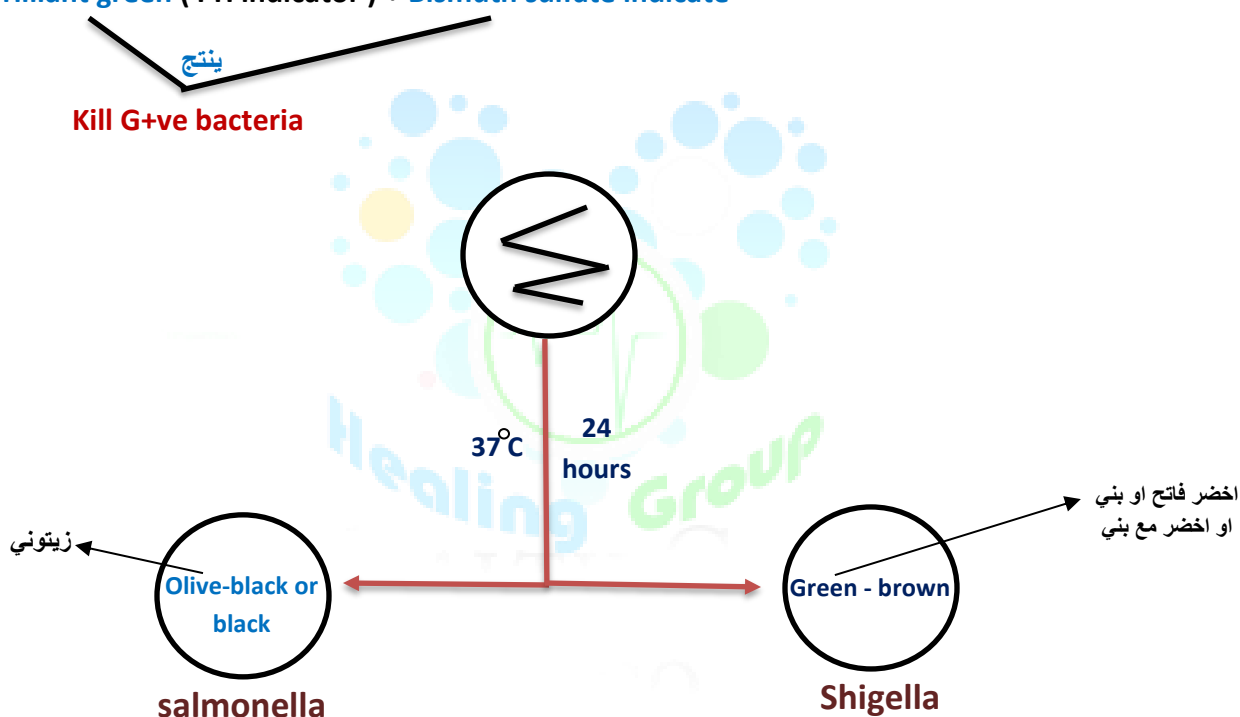
- *mannitol → sugar
- *NaCl (7.5%) → salt
- *PH indicator (phenol red) → neutral → red
Acid → yellow
Basic → pink

* halophiles → salt-loving bacteria.
↓
Such as staphylococcus.

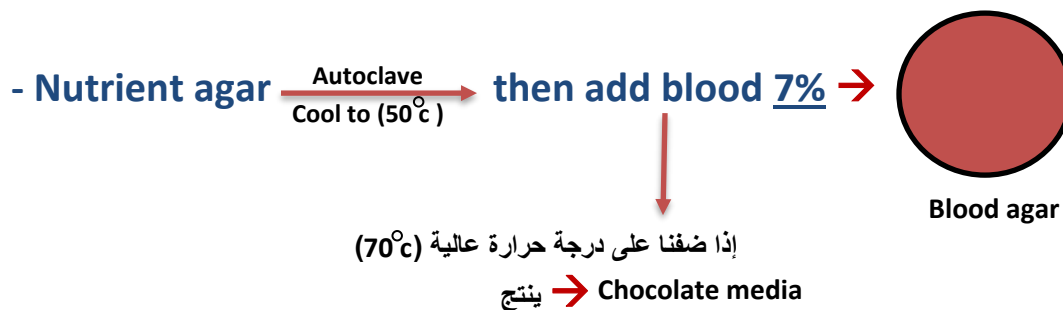


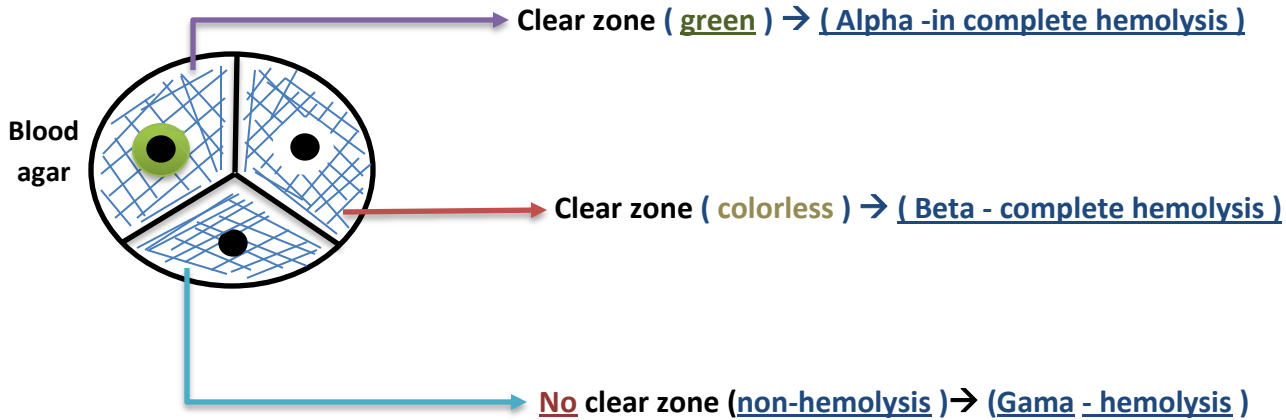
6- Bismuth :

- * enzymatic digest casein
- * enzymatic digest Animal tissue
- * Dextrose → C-source
- * Bismuth sulfate indicate → (sulfur source) $\xrightarrow{\text{Bacteria degrade}}$ $\text{H}_2\text{S}_{(g)} \rightarrow \text{colorless}$
+
 $\text{Fe}^{+2} \rightarrow \text{FeS (Black)}$
- * Ferrous sulfate → Sulfur indicate .
- * Brilliant green (PH indicator) + Bismuth sulfate indicate



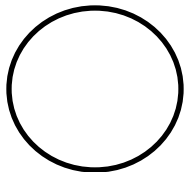
*Enrichment – differential (blood agar)





Clear zone → clear of blood

*General – differential (Nutrient agar)



Media surrounding colony **green** (pseudomonas aeruginosa) → **pyocyanin** لأنها تحتوي على طبقة

please forgive us for any mistake!

Healing group " Nursing group "



تم بحمد الله

Sheet

What is Media?

Bacteria grown in a laboratory environment, like captive animals in a zoo, need to have everything provided for them—food, water, a suitable environment to survive and thrive. Some microbes are not especially choosy in their requirements for growth, while others, such as *Treponema pallidum*, the causative agent of syphilis, has never been successfully grown in culture, although scientists have been trying to do so for more than 100 years.

Several basic types of media are discussed below. Although their differences are featured, there are several characteristics that all culture media have in common. Media must be prepared in such a way that it is sterile prior to being inoculated with a bacterial sample, so that when a particular type of bacteria is cultured (cultivated) on that medium, it is the only type of bacteria present. Growth media must also provide everything the bacterial culture needs to live and grow, including water, nutrients, and the proper pH. Media can be either liquid (nutrient broth) or solid (agar).

Defined Media versus Complex Media

Some media formulations are very specific recipes in which certain ingredients must be present in specific amounts. These defined media (also known as synthetic media) are used to grow bacteria that have very particular needs.

Most clinical cultures do not have such exacting requirements, and can be grown in what is referred to as “complex media”. Complex media are composed of partially digested yeast, beef, soy and additional proteins, in which the exact concentration and composition is unknown. In comparison with defined media, which are good for growing bacteria with very particular needs, complex media can be thought of as a crowd-pleaser, suitable for growing many different types of less fastidious microbes.

In addition to growth media formulations being classified as either defined or complex, there are also media that are designed to do more than just grow bacteria, **selective and differential media provide information about the bacteria growing.**

Type of the selective media according to it's purposes:

1. **General media:** grow on it most of the bacteria. E.g. Nutrient agar and nutrient broth
2. **Selective media :** Selective media contain ingredients that inhibit the growth of certain types of bacteria and/or encourage the growth of others. This type of media is useful in helping to identify unknown bacteria and in encouraging the growth of only the types of bacteria that the microbiologist is interested in cultivating. g Mannitol salt agar for *Staphylococcus* and *Pseudomonas* F agar for *Ps. Aeruginosa*
3. **Differential media:** A differential medium distinguishes between different types of bacteria based on some characteristic of the bacteria that is growing on it. Typically there is a color change that results from certain bacterial metabolic products reacting with substances or chemicals that have been added to the media.

Differential culture media are formulated to display a color change when the bacteria growing utilize a certain ingredient. **Blood agar media** which differentiate between Beta-, alpha-, and Gama- Hemolysis

4. **Selective differential media :** Some specialized media are both selective and differential, others are either or. The MacConkey's and Mannitol Salt described above are both selective and differential bacterial growth media. Also, EMB and TSI

5. **Enrichment media:** Rich with nutrient that able for all bacteria to grow on it

with one more

6. **Enrichment differential media:** select the bacteria and differentiate from

the other type of bacteria

e.g Blood agar.

- **Nutrient agar:** is the general media all for most type of bacteria to grow on it, but some time may be use to differentiate between types of bacteria such as *Pseudomonas aeruginosa* which give green pigment on nutrient agar (*Pyocyanin*)

-MacConkey's Agar:

Components:

- * **Bile salts** - Inhibits most Gram-positive bacteria, except *Enterococcus* and some species of *Staphylococcus* i.e. *Staphylococcus aureus*.
- * **Crystal violet dye**- Inhibits certain Gram-positive bacteria thus selecting for Gram negatives.
- * **Lactose**- Some bacteria can ferment lactose acid-end products, others cannot.
- * **Neutral pH red indicator** - Stains microbes fermenting lactose
- * **Peptone** - a source of proteins, amino acids for microbial growth.

For example (MAC) is used to cultivate Gram-negative bacteria, by discouraging the growth of Gram positive bacteria through the use of crystal violet dyes and bile salts . MAC has special additives that cause lactose fermenting bacteria (microbes that can metabolize the sugar lactose) to grow in pink colonies, whereas Gram-negative non-lactose fermenting bacteria will grow in colorless colonies.

If the Gram- organisms growing on **MacConkey's Agar** eat lactose, they produce acidic metabolites (wastes) that trigger the neutral red pH indicator. This causes the bacterial colonies themselves to turn mauve or pink in color.

EMB:

Components:

- **Lactose** – a disaccharide which can be fermented by some bacterial **enzymes to produce acidic end products**.
- **Eosin and Methylene Blue** – these are dyes which inhibit the growth of most **Gram positive** bacteria. They also react with any acidic products resulted from lactose fermentation to color the colonies.

Acid production from lactose fermentation causes precipitation of the dyes on the surface of the colony resulting in different colors.

- **Large amounts of acid** → green metallic sheen
- **Small amounts of acid** → pink
- **No fermentation** → colorless

Enterobacter aerogenes produces large colonies which are pink-to-buff around dark centers. *Escherichia coli* produce small, dark colonies with a green metallic sheen. *Pseudomonas*, *Proteus*, *Salmonella* and *Shigella* sp produces colorless colonies because it does not ferment lactose.

Eosin methylene blue agar is a differential medium used for the detection and isolation of Gram-negative intestinal pathogens. A combination of eosin and methylene blue is used as an indicator and allows differentiation between organisms that ferment lactose and those that do not. Sucrose is also included in the medium because certain members of the *Enterobacteria* or coliform group

ferment sucrose more readily than they ferment lactose. In addition, methylene blue acts as an inhibitor to

Gram positive organisms. Colonies of *E. coli* normally have a dark center and a greenish metallic sheen, whereas the pinkish colonies of *Enterobacter aerogenes* are usually mucoid and much larger than colonies of *E. coli*. Other organisms, such as *Salmonella* (one of the causative agents of food poisoning), do not ferment lactose or sucrose and produce colonies that are non colored.

XLD:

Xylose lysine deoxycholate agar (XLD agar) is a selective growth medium used in the isolation of Salmonella and Shigella species from clinical samples and from food.^{[1][2]} It has a pH of approximately 7.4, leaving it with a bright pink or red appearance due to the indicator phenol red. Sugar fermentation lowers the pH and the phenol red indicator registers this by changing to yellow. Most gut bacteria, including *Salmonella*, can ferment the sugar xylose to produce acid; *Shigella* colonies cannot do this and therefore remain red. After exhausting the xylose supply *Salmonella* colonies will decarboxylate lysine, increasing the pH once again to alkaline and mimicking the red *Shigella* colonies. *Salmonellae* metabolize thiosulfate to produce hydrogen sulfide, which leads to the formation of colonies with black centers and allows them to be differentiated from the similarly colored *Shigella* colonies.

Other *Enterobacteria* such as E. coli will ferment the lactose and sucrose present in the medium to an extent that will prevent pH reversion by decarboxylation and acidify the medium turning it yellow.

- *Salmonella* species: red colonies, some with black centers. The agar itself will turn red due to the presence of *Salmonella* type colonies.
- *Shigella* species: red colonies.
- Coliforms: yellow to orange colonies.
- *Pseudomonas aeruginosa*: pink, flat, rough colonies. This type of colony can be easily mistaken for *Salmonella* due to the color similarities.

XLD agar contains:

Yeast extract	3g/l
L-Lysine	5g/l
Xylose	3.75g/l
Lactose	7.5g/l
Sucrose	7.5g/l
Sodium deoxycholate	1g/l
Sodium chloride	5g/l
Sodium thiosulfate	6.8g/l
Ferric ammonium citrate	0.8g/l
Phenol red	0.08g/l
Agar	12.5g/

Hektoen agar:

Components:

- **Bile salts:** Inhibits the growth of most Gram positive organisms.
- **Bromthymol blue and acid fuchsin dyes:** have a lower toxicity than that of many other enteric media, resulting in improved recovery.
- **Lactose, sucrose, and salicin:** provide fermentable carbohydrates to encourage the growth and differentiation of enterics.
- **Sodium thiosulfate:** provides a source of sulfur.
- **Ferric ammonium citrate:** provides a source of iron that allows the production of hydrogen sulfide from sodium thiosulfate, which provides a source of sulfur. This also allows the visualization of hydrogen sulfide production by reacting with hydrogen sulfide to form a black precipitate

Type: **Selective and differential**

Purpose: Detects lactose fermentation, H₂S production, inhibits non-enteric

Interpretation: Lactose fermenters yellow or salmon, non-fermenters colorless; H₂S production produces black precipitate

Mannitol salt agar:

Components:

- **7.5% NaCl** – selects for species of *Staphylococcus*.
This concentration of salt is too high for most other bacteria to withstand and therefore, inhibits their growth.
- **Mannitol** – alcohol of the carbohydrate mannose.
Mannitol fermentation produces acid end products which turn the medium yellow.
Yellow indicates mannitol positive and no color change indicates mannitol negative.
- **Phenol red pH indicator** – yellow in acid pH
(The same indicator that is used in phenol red carbohydrate fermentation broths).

(MSA) has a high concentration of sodium chloride, which selects for halophiles (salt-loving bacteria) such as members of the genus *Staphylococcus*.

Mannitol Salt Agar also contain **food (mannitol, a sugar alcohol) and a pH sensitive dye**. When the bacteria growing on MSA ferment mannitol, the medium changes from its original pink color to a bright yellow. Another specialized medium

This medium only grows salt-loving bacteria (so it is selective). In addition, bacteria that grow on MSA that can ferment mannitol, a sugar alcohol, will turn the medium from its original pink color to a bright, neon yellow. This color change is clinically significant in that *Staphylococcus aureus*, a disease-causing bacterium, is a mannitol fermenter, while *S. epidermidis*, a beneficial bacteria that normally grows on the surface of the skin and mucous membranes, is not a mannitol fermenter.

Blood Agar (BAP) contains sheep's blood, if bacteria growing on the medium produce exotoxins that hemolyze (cut up blood cells), the medium changes color.

This high-nutrient growth medium is made with 5% sheep's blood. **Blood Agar** is not selective, and will grow a wide range of bacteria. This medium is, however, differential.

Some bacteria produce hemolysins, exotoxins that can break down red blood cells (hemo = blood and lyse = to cut). Depending on the type of hemolysin, the red blood cells are either completely broken down (beta-hemolysis), or partially broken down (alpha-hemolysis).

Streptococcus pyogenes is a beta-hemolytic pathogenic bacterium and the causative agent of Strep throat. When plated on Blood Agar, *S. pyogenes* completely breaks down the red blood cells, leaving the areas of agar around its colonies clear and colorless.

Most normal throat flora either produce alpha-hemolysis, partially breaking down the red blood cells and creating a bruised appearance to the agar, or produce gamma-hemolysis (which means no hemolysis at all), leaving the agar is unaffected.

Bismuth Sulfite Agar

Summary and Explanation Salmonellosis continues to be an important public health problem worldwide, despite efforts to control the prevalence of *Salmonella* in domesticated animals. Infection with nontyphi *Salmonella* often causes mild, self-limiting illness.¹ Typhoid fever, caused by *S. Typhi*, is characterized by fever, headache, diarrhea and abdominal pain, and can produce fatal respiratory, hepatic, splenic and/or neurological damage. These illnesses result from consumption of raw, undercooked or improperly processed foods contaminated with *Salmonella*. Many cases of *Salmonella*-related gastroenteritis are due to improper handling of poultry products. United States federal guidelines require various poultry products to be routinely monitored before distribution for human consumption but contaminated food samples often elude monitoring. Bismuth Sulfite Agar is a modification of the Wilson and Blair²⁻⁴ formula. Wilson^{5,6} and Wilson and Blair²⁻⁴ clearly showed the superiority of Bismuth Sulfite medium for isolation of *S. Typhi*. Cope and Kasper⁷ increased their positive findings of typhoid from 1.2 to 16.8% among food handlers and from 8.4 to 17.5% among contacts with Bismuth Sulfite Agar. Employing this medium in the routine laboratory examination of fecal and urine specimens, these same authors⁸ obtained 40% more positive isolations of *S. Typhi* than were obtained on Endo medium. Gunther and Tuft,⁹ employing various media in a comparative way for the isolation of typhoid from stool and urine specimens, found Bismuth Sulfite Agar most productive. On Bismuth Sulfite Agar, they obtained 38.4% more positives than on Endo Agar, 33% more positives than on Eosin Methylene Blue Agar, and 80% more positives than on the Deoxycholate media. These workers found Bismuth Sulfite Agar to be superior to Wilson's original medium. Bismuth Sulfite Agar was stable, sensitive and easier to prepare. Green and Beard,¹⁰ using Bismuth Sulfite Agar, claimed that this medium successfully inhibited sewage organisms. The value of Bismuth Sulfite Agar as a plating medium after enrichment has been demonstrated by Hajna and Perry.¹¹ Since these earlier references to the use of Bismuth Sulfite Agar, this medium has been generally accepted as routine for the detection of most *Salmonella*. The value of the medium is demonstrated by the many references to the use of Bismuth Sulfite Agar in scientific publications, laboratory manuals and texts. For food testing, the use of Bismuth Sulfite Agar is specified for the isolation of pathogenic bacteria from raw and pasteurized milk, cheese products, dry dairy products, cultured milks and butter.^{1,12-14} The use of Bismuth Sulfite Agar is also recommended for use in testing clinical specimens.

15,16 In addition, Bismuth Sulfite Agar is valuable when investigating outbreaks of *Salmonella* spp., especially *S. Typhi*.17-19 Bismuth Sulfite Agar is used for the isolation of *S. Typhi* and other *Salmonella* from food, feces, urine, sewage and other infectious materials. The typhoid organism grows luxuriantly on the medium, forming characteristic black colonies, while gram-positive bacteria and members of the coliform group are inhibited. This inhibitory action of Bismuth Sulfite Agar toward gram-positive and coliform organisms permits the use of a much larger inoculum than possible with other media employed for similar purposes in the past. The use of larger inocula greatly increases the possibility of recovering the pathogens, especially when they are present in relatively small numbers. Small numbers of organisms may be encountered in the early course of the disease or in the checking of carriers and releases. Principles of the Procedure In Bismuth Sulfite Agar, beef extract and peptone provide nitrogen, vitamins and minerals. Dextrose is an energy source. Disodium phosphate is a buffering agent. Bismuth sulfite indicator and brilliant green are complementary in inhibiting gram-positive bacteria and members of the coliform group, while allowing *Salmonella* to grow luxuriantly. Ferrous sulfate is included for detection of H₂S production. When H₂S is present, the iron in the formula is precipitated, giving positive cultures the characteristic brown to black color with metallic sheen. Agar is the solidifying agent.

* Formula Difco™ Bismuth Sulfite Agar Approximate Formula* Per Liter

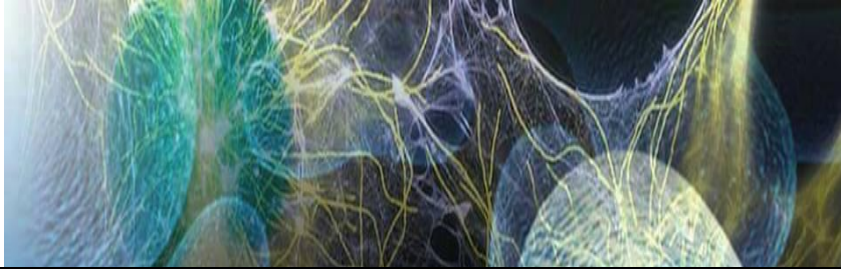
Beef Extract	5.0 g
Peptone	10.0 g
Dextrose	5.0 g
Disodium Phosphate	4.0 g
Ferrous Sulfate	0.3 g
Bismuth Sulfite Indicator	8.0 g
Agar	20.0 g
Brilliant Green.....	25.0 mg

Results:

Typical *S. typhi* surface colonies are black, surrounded by black or brown-black zone with or without a metallic sheen. This zone may be several times the size of the colony. Other strains of Salmonella produce black to green colonies with little or no darkening of surrounding medium. Shigella spp. other than *S. flexneri* and *S. sonnei* are inhibited. *S. flexneri* and *S. sonnei* strains that do grow on this medium produce brown to green, raised colonies with depressed centers and exhibit a crater-like appearance.

Healing Group

وعزمنّا أن نحسن سعيّا



Biochemical tests (Final)

2017

Microbiology Lab



Done by: Mohamad BshaiRi

كلية التمريض



Biochemical tests



Metabolic enzymes

Endoenzyme (Intracellular)

Work inside cell

Exoenzyme (Extracellular)

Work in media

* Intracellular enzyme (**Carbohydrate Fermentation test**):

Media TSI (triple sugar iron agar)

* 3 sugar → glucose, Lactose, sucrose → (C-source)

* Peptone (protein source).

* PH indicator (**phenol red**) → neutral → **Red**
Acid → **yellow**
Basic → **Pink**

* Sulfur source → Sodium thiosulfate or **cysteine**

* Sulfur indicator → ferrous ammonium citrates

ملاحظة :

إذا كان في سؤال بالامتحان

يسأل عن Sulfur source وكان بالخيارات

cysteine راح يكون هو الجواب ولو كان في خيار فيه

Sodium thiosulfate or cysteine

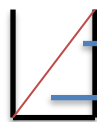
راح يكون هو الجواب بشرط ما يكون في احد

الخيارات لحاله cysteine

وإذا ما كانت بالخيارات بنختار الصوديوم

Amino acid

TSI → (slant media)



Slant

butt

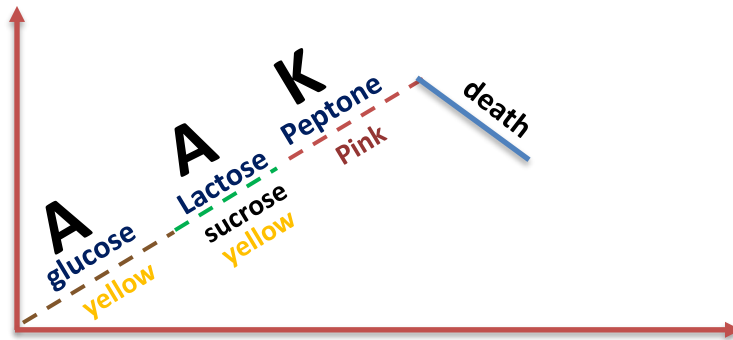
By loop make streaking (gram negative)

By needle make stabbing (gram negative)

Same bacteria

* Incubate for not more 18 hours

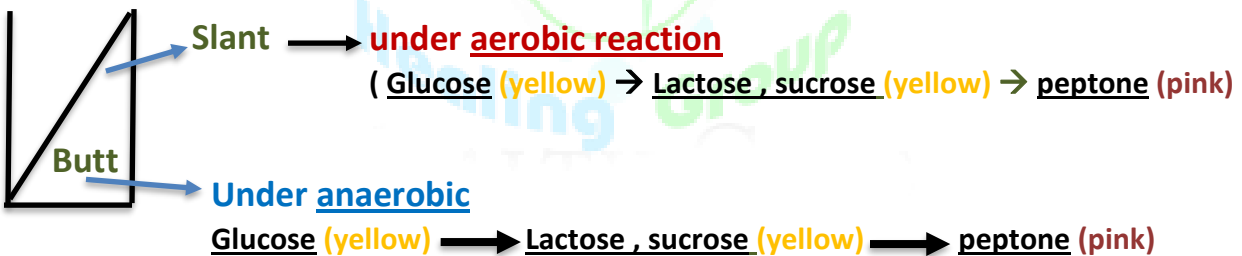
لأنه سكر بكفي ١٨ ساعة وبعدها بصير بكسر الـ **Peptone**



* **Black color** → **H₂S producer**
 colorless
 $\text{Fe}^+ \rightarrow \text{FeS (black)}$ **Only in butt**

الغاز بالـ Slant معه مجال يطلع
 اما في الـ butt ما في مجال يطلع
 ف بتفاعل مع ferrous ammonium
 وبعطي لون أسود

* **تشققات** cracking → **only in butt** → Make by gas (mean gas producer)



(Reaction in different condition)

Aerobic faster than **anaerobic**

الي بكسر الـ lactose بكسر كل أنواع السكر

*Results:

Color of slant

Color of butt

$\frac{\text{Yellow}}{\text{Yellow}} = \frac{A}{A}$ (Lactose – fermenter)

- Crakes $\rightarrow \frac{A}{A}$ (Lactose-fermenter with gas producer)

It mean gas producer

- $\frac{A}{A}$ $\rightarrow$ (Lactose-fermenter and H₂S(g) producer)
Black in butt $\leftarrow \frac{A}{A} \cdot H_2S$

- $\frac{\text{Pink}}{\text{Yellow}} = \frac{K}{A}$ $\rightarrow$ (Non-Lactose ferment) $\xrightarrow{\text{توضيح}}$
قبل ١٨ ساعة

- $\frac{K}{A}$ **crakes** (Non-Lactose fermenter with gas producer)

- $\frac{K}{A} \cdot H_2S$ $\rightarrow$ (Non-Lactose fermenter with H₂S(g) producer)

الـ slant بس تخلص من glucose
بدھا enzyme حتى تكسر lactose and sucrose
بس ما في هذا الـ enzyme
فيبتروح على الـ peptone
اما الـ butt لساتھا بالخطوة الاولى
مشان هيك A

All these are carbohydrate fermenter bacteria

- $\frac{K}{K}$ → can degrade peptone under aerobic and anaerobic

- $\frac{K}{(K)}$

- $\frac{K}{(K).H_2S}$

These are Non-carbohydrate fermenter

* $\frac{K}{\text{Red}}$ → Non-carbohydrate degrade peptone only under aerobic

↓
No change
No crakes
No black

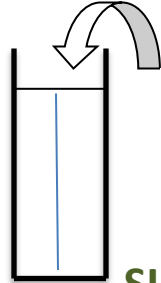
* $\frac{K}{\text{Black}}$ → if the black color mask butt color you should consider butt as yellow

↓
 $\frac{K}{A}$

* $\frac{A}{K}$ → wrong result because slant faster than butt

IMvic test

* Indol – test



Stabbing by needle single line then incubate 24 hours at 37 °c

SIM (sulfide indol motility)

Black color

(sulfide +)

no change color

(sulfide-)

growth in all media

(motility+)

growth only in inoculated line

(motility-)

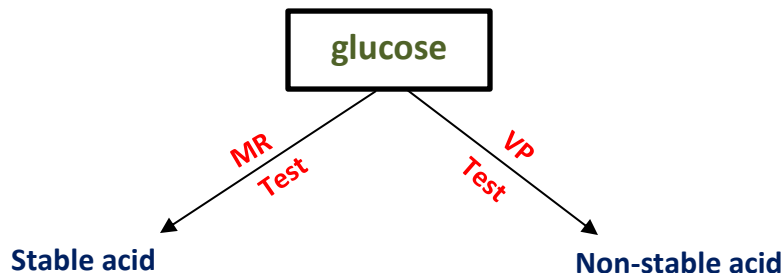
* Then add **Kovac's reagent** and wait for 10 min

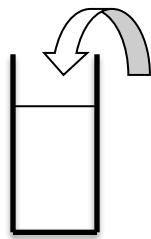
- red ring → indol positive

- No change (other color except red) → indol negative

إذا ما طلع red ring بتستنى ١٠ دقائق
لحتى يتحول اللون
وإذا ما طلع خلال ١٠ دقائق
يكون indol negative

* MR-VP test → methyl red- voges proskauer

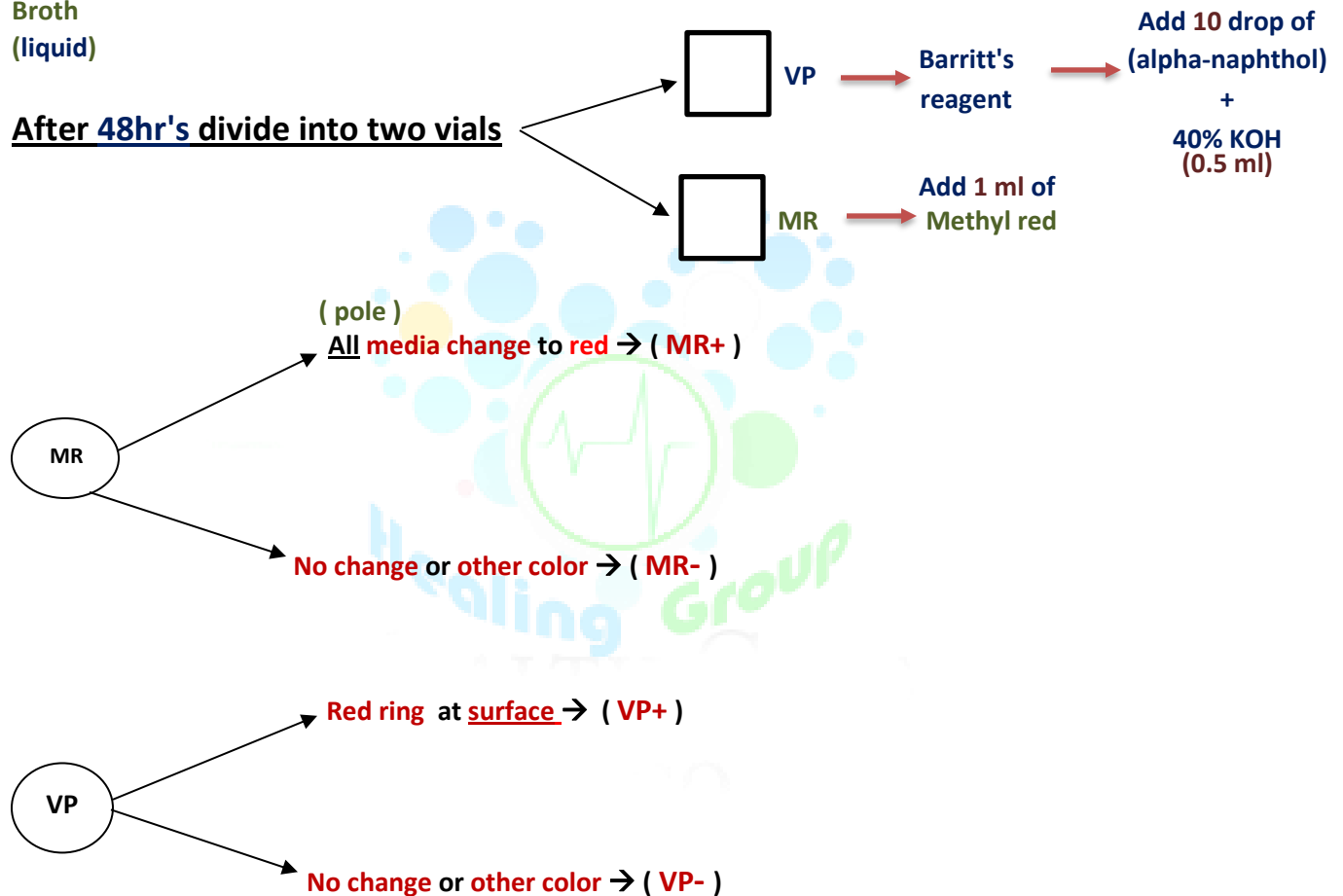




By loop add bacteria and mix well, then incubate for 48 hours

MR-VP
Broth
(liquid)

After 48hr's divide into two vials



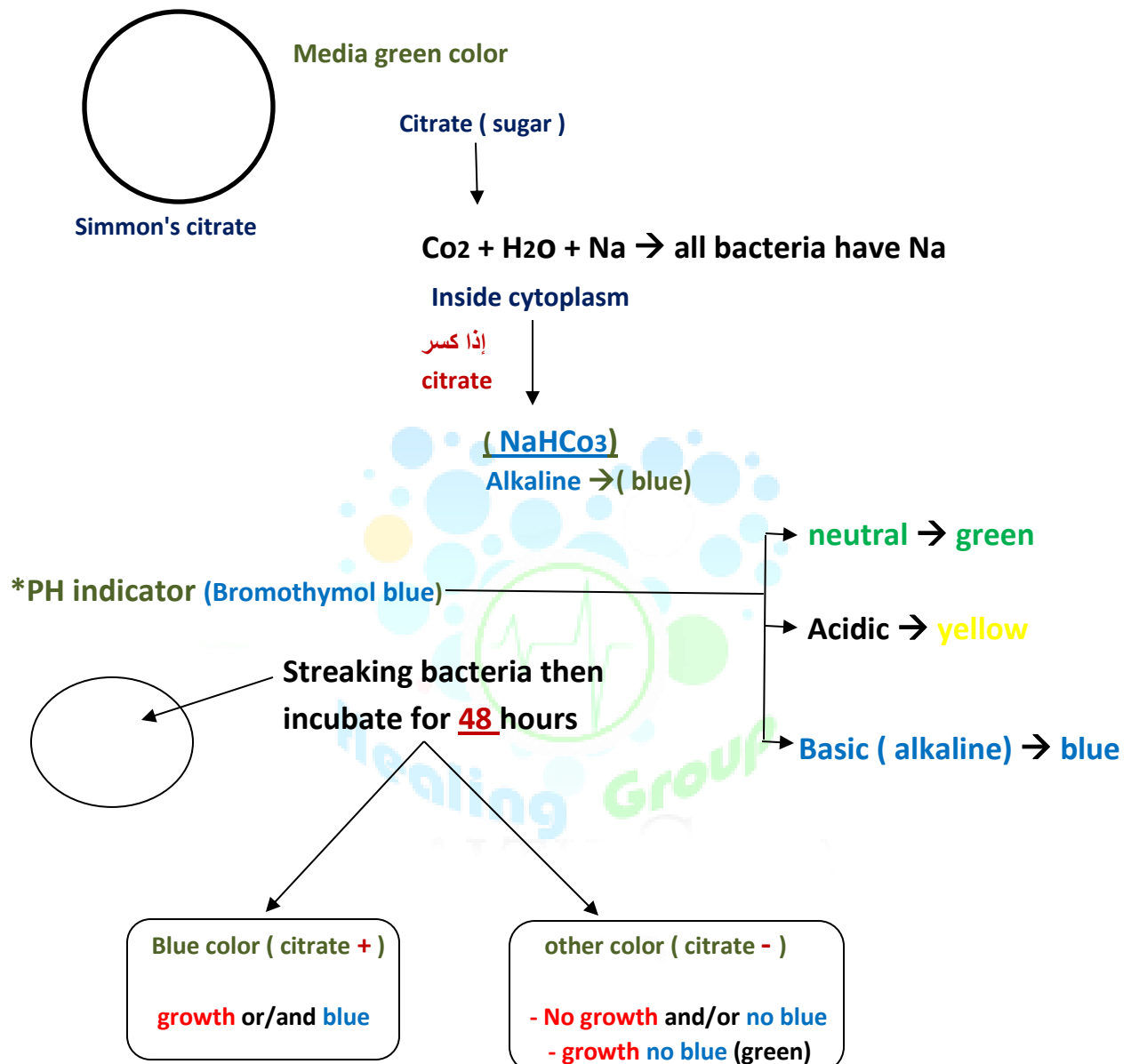
MR+ , VP+ → wrong (مستحيل تطلع النتيجة هيك)

MR+ , VP- → stable acid

MR- , VP+ → unstable acid

MR- , VP- → carbohydrate Non-fermenter (K/K)

* Citrate test:



*catalase test → biochemical test

ارجع لل O2 requirement test



slide

- * add one drop of H_2O_2
- * add loopful of bacteria then mix

Source of enzymes

* show air bubbles

* No air → catalase (-)

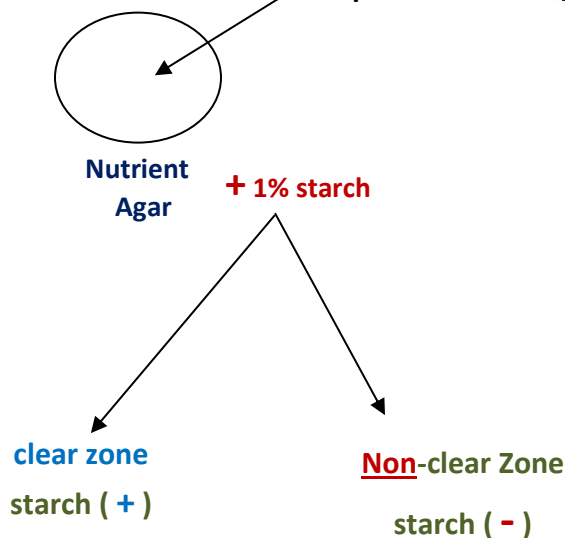
* large amount of air bubbler → strong catalase (+)

* medium or small air bubbler → weak catalase (+)

*exoenzymes:

(1) starch test → polymer of glucose

Loopful of bacteria in the center incubate for 48 hours

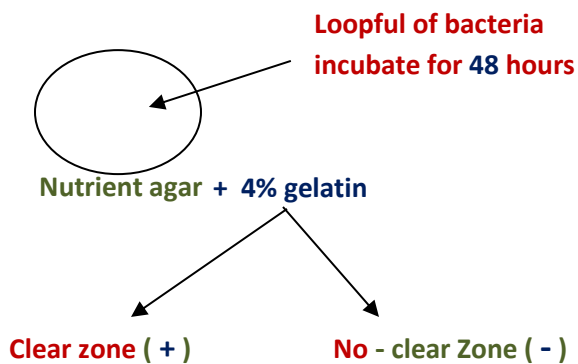


* Use indicator of starch → iodine

- starch + I_2 → blue-black around colony (-)

- starch + I_2 → brown around colony (+)

(2) gelatin test → polymer of amino-acid



Gelatinase enzyme → break of gelatin

Colorless

Hgcl₂/ HCl + gelatin → white around colony (All media → -)

(+) → All media white except around colony (+)

please forgive us for any mistake!

Healing group " Nursing group "



تم بحمد الله

Biochemical Tests

Classification of bacteria: all of the following are needed in classification of bacteria: •

Macroscopic characteristic, includes:

Colony morphology.

- Growth morphology

• Microscopic characteristic, includes: -

Cell size

- Cell shape

- Cell arrangement

- Endospore staining

- Acid-fast staining

• Physical factors:

- Temperature

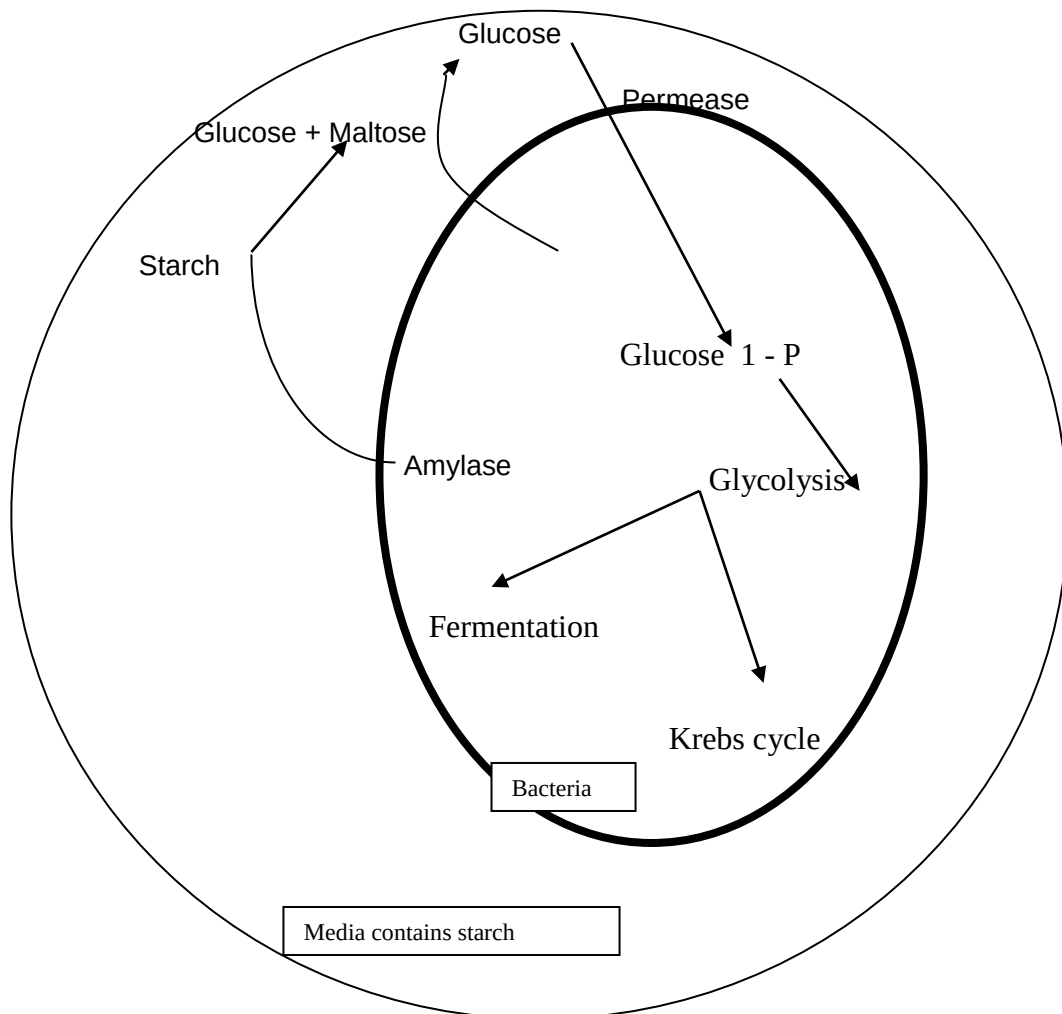
- pH

O₂ requirements

- Osmotic pressure

• Biochemical test:

Each bacteria has which is called biochemical fingerprints which is a properties controlled by the cells' enzymatic system, these includes bioenergetics, biosynthesis, and biodegradation. These pathways called cellular metabolism. The enzyme which controlled the cellular metabolism divide according to where it's work into two types, these are Extracellular and intracellular enzymes.

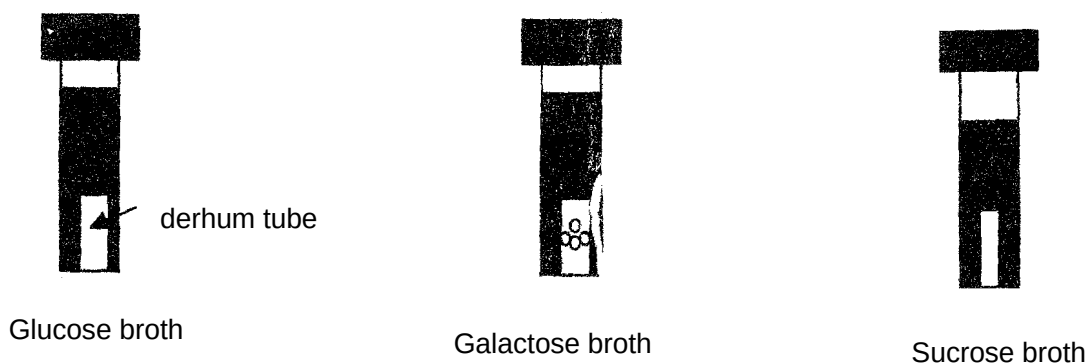


Amylase and *Maltase* are extracellular enzymes which are synthesized in bacteria and secreted outside the cell to convert high molecular compounds to low molecular ones (complex to simple). All enzymes involved in fermentation, glycolysis, and Krebs cycle are intracellular enzymes.

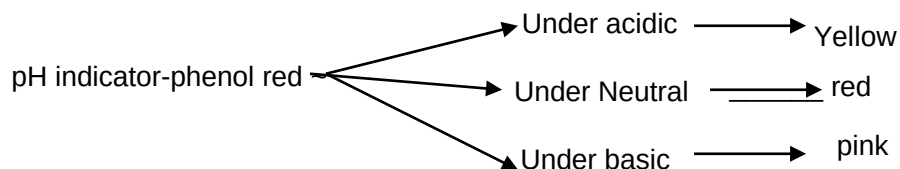
INTRACELLULAR ENZYMES

** Carbohydrate fermentation : Achieved by one of the following pathways:

- a. Pentose shunt
- b. Entner Doudoroff
- c. Embden Meyerhof.



Each test tube contains in addition to a single carbohydrate the following:



- Derhum tube: for gas collection
- Nutrient agar: support the growth of bacteria.

Inoculate each test tube with a loopful from your plate by sterile cooled loop, then mix gently with loop avoid trapping the air bubble in derhum tube, and incubate at 37°C for 24 hours then check the color of the media and if there is Gas in the Derhum test tube or not.

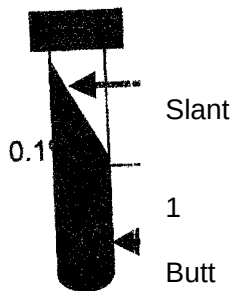
What is the difference between respiration and fermentation?

Respiration : either Aerobic:
Anaerobic:

in biooxidations use O₂ as a final electron acceptor. in biooxidations use inorganic ions as a final electron acceptor e.g. NO₃⁻ or SO₄⁼

Fermentation:

in biooxidations use organic substances as a final electron Acceptor and need no oxygen.



pH indicator : phenol red

carbohydrate: lactose and sucrose 1.0%, and

glucose 0.1%

** Triple sugar iron agar (TSI): for gram negative bacteria only

This media contains:

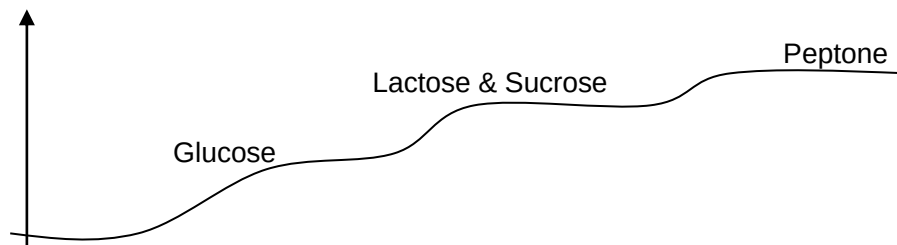
pH indicator- Phenol red

Sodium thioglycolate or cysteine: sulfur source

Ferrous sulfate (FeSO_4) : Sulfur indicator

Peptone: carbon source

Bacteria if have enzymes to degrade both glucose, sucrose, lactose and peptone they start with the simplest then more and more complex as in the following curve:



Inoculation:

By sterile cooled needle inoculate the butt of the TSI at are 1" from your pure culture. Then from the same culture inoculate the surface by sterile cooled loop through streaking. Incubate at 37°C for not more than 18 hours.

Results

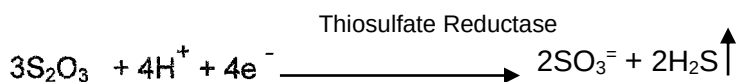
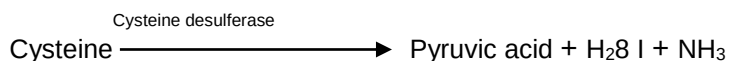
Slant / Butt → Yellow / Yellow → Acid / Acid → A / A

Lactose and / or sucrose ferment or . If this bacteria inoculated on MacConkey it will gave a pink colonies, but on EMS It will give colored colonies. If there is cracking

in the butt, there is gas produced, the result A/A , if there is black color in the butt or between the butt and the slant, 5thi~ria are H_2S producer, the

result is A/A , H_2S

This bacteria have enzymes to ferment the lactose or / and sucrose which are found in high concentration. Also this bacteria decompose cysteine or sodium thiosulfate as followings:



H₂S is colorless gas, dissolve in water. This gas will react with FeSO₄ (H₂S indicator) as following:



OR

The result is Pink / Yellow $\longrightarrow$ Alkaline/ Acid $\longrightarrow$ K / A,
 K/Ⓐ , K/Ⓐ , H₂S

This bacteria ferment glucose first, and because this substrate found in small concentration the small acid which produced in the slant oxidized and the peptone is degrade to produce ammonia which raise the pH toward the alkaline condition Pink color.

OR

The result is Pink! Pink $\longrightarrow$ Alkaline/ Alkaline $\longrightarrow$ K/K
 K/Ⓐ , K/Ⓐ , H₂S

There is no carbohydrate fermentation, instead, peptone catabolized under aerobic and anaerobic condition, resulting in alkaline pH due to production of ammonia.

OR

The result is Pink / red $\longrightarrow$ Alkaline/nochange $\longrightarrow$ K/ no change
 only peptone catabolize under anaerobic condition.

If the black color mask the color of butt, the color of the butt is considered yellow.

Why the incubation must be not longer than 18 hours?

Because for example the result is A / A and the bacteria incubate longer than 18 hours, the bacteria ferment lactose and sucrose completely, then start to degrade peptone which change the pH of the media from acid to alkaline and the result will appear K / K.

Why there is no A / K?

Because the reaction under aerobic condition is faster than that under anaerobic.

AEROBIC:

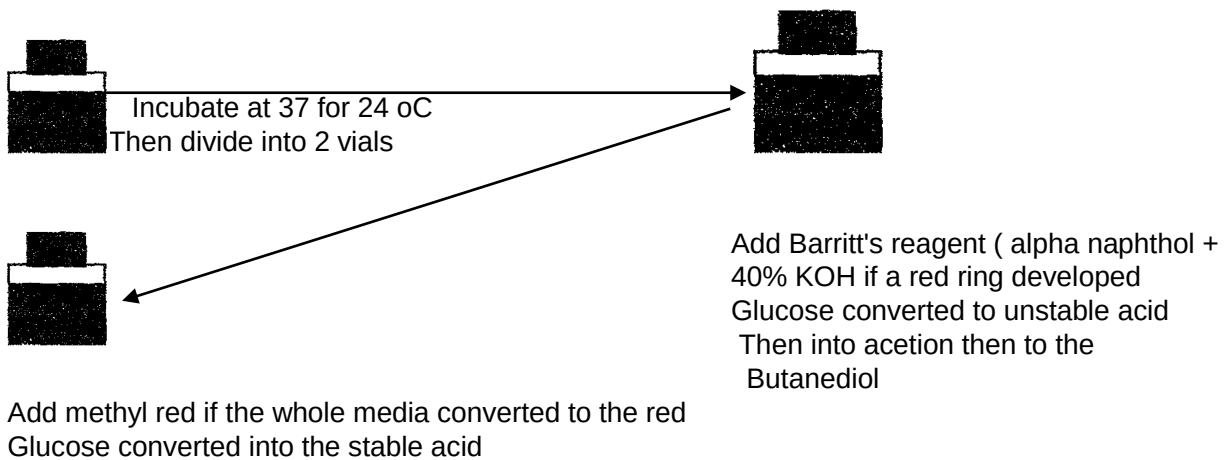


ANAEROBIC:

** Indole: Add about one rnl of kovac's reagent to the media if a red ring developed within 10 minutes after the addition the bacteria degrade the Tryptophan to the indole, pyruvic acid, and Ammonia by Tryptophanase.

Carbohydrate $\xrightarrow{\text{x3}}$ Acid (yellow) $\xrightarrow{\text{x4}}$ Alkaline (Pink)

Inoculate this media by a loopful of your pure culture using steriled cooled loop and incubate at 37°C for 48 hours. After

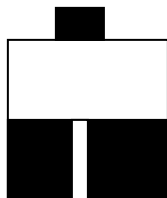


The result must be

If the reaction under anaerobic at X3 the reaction under aerobic must be at X2, If the reaction under anaerobic at X4 the reaction under aerobic must be already completeid.

IMViC; TEST:

I: for indole, M : test, methyl red, Vi : for voges proskauer
C: for Citrate.



Inoculate this **SIM** media from your culture by just single line inoculation Using steriled cooled needle, then incubate at 37°C for 48 hours.

S : for sulfide, **I** : for Indole, and **M**: for Motility.

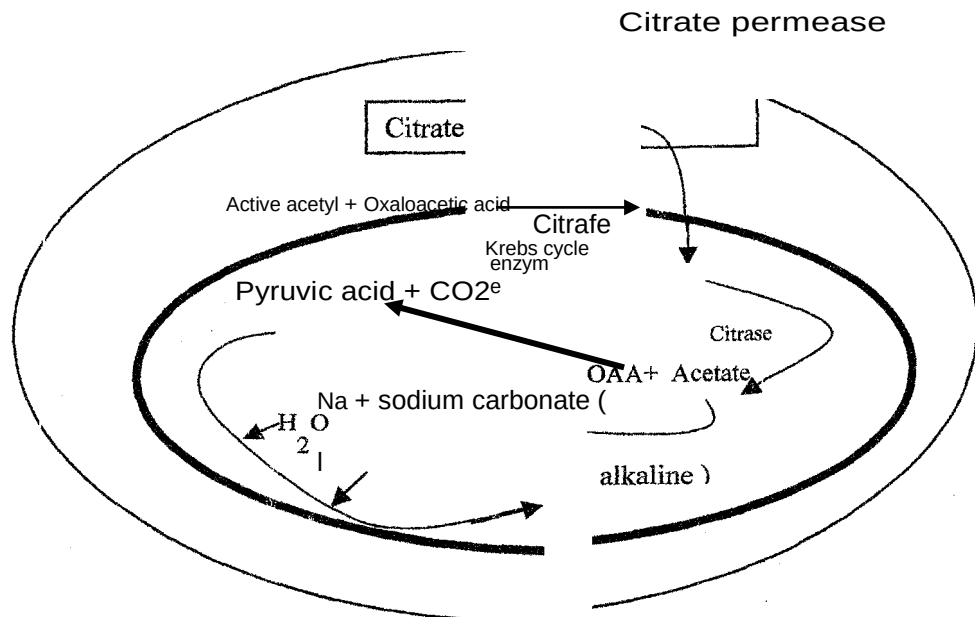
Results: take the result in the following order:

** Look at the presence of black color, Black color indicates for sulfur production as in TSI.

** Motility: if the bacteria grow at the inoculating line only the bacteria is not motile.

If the bacteria grow at the surface of the media, or the bacteria grow a way from inoculating line, or the media change holly to the black color the bacteria is motile.

MR - VP : MR : for Methyl red, VP : for Voges proskauer test



Urease test:



Inoculate the urea agar slant with a loopful of your pure culture then incubate at 37°C for 24 to 48 hours then check for the presence of pink color. After 24 hours if all the media converted completely to pink the bacteria is strong urease positive, if the media is partially converted to pink the bacteria is weak urease positive. No pink color the bacteria is urease negative.

The bacteria which has a positive urease has this hydrolytic enzyme which attacks the nitrogen and carbon bond in amide compounds such as urea to form CO₂ + water + Ammonia. Ammonia raise the pH (alkaline condition) and the phenol red convert

the media into pink color.

Example: *Proteus*.

MR : +ve must be VP : -ve OR MR: -ve must be VP : +ve
 OR MR: -ve must be VP: -ve
 But impossible to be : MR: +ve VP: +ve

both

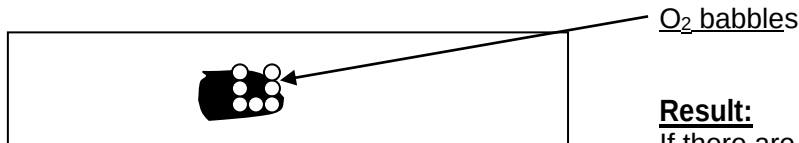
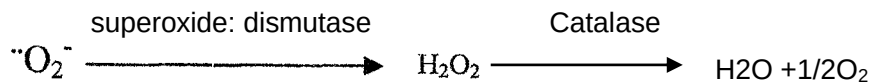
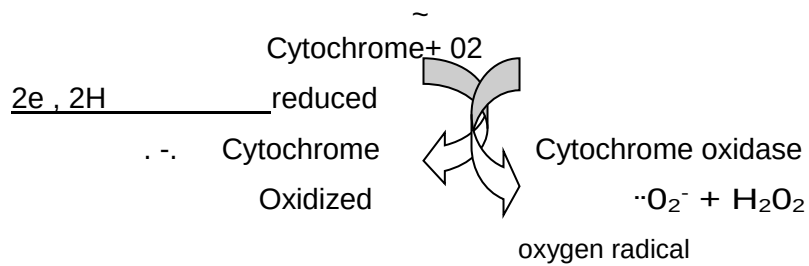
substances are toxic for bacteria, so the above mentioned group of bacteria have superoxide dismutase and catalase enzymes. pH indicator: is bromothymol blue: yellow under acidic, Green under neutral, and blue under alkaline
 Super oxide dismutase

Citrate test: ..

Inoculate the surface of simmon's citrate media with a loopful of your pure culture using sterilized cooled loop, then incubate at 37°C for 48 hours. If there is growth and/or the color of the media converted from green to the blue, this indicates that the bacteria use citrate as a carbon source. The following is occur:

Catalase and oxidase:

In aerobic, facultative anaerobic, and anaerobic aerotolerant use O_2 as final electron acceptor



Result:

If there are large amount of bubbles

Small amount of bubbles
No bubbles

- 1- Add loopful from your pure culture into cleaned slide.
- 2- Add 3-5 drops of 3% H_2O_2 (substrate)

Strong catalase positive Weak
catalase positive Catalase
negative

Oxidase test:

Reveal the presence of Cytochrome oxidase

Test:

Either flood the plate of bacterial culture with a solution of p - aminodimethylaniline oxalate if the pink, then maroon, and finally black coloration developed at the surface of colonies is indicative of Cytochrome oxidase

Production (catalase positive), others is catalase negative. **OR** cut the filter paper into small pieces then immerse them in 0.001% P - aminodimethylaniline oxalate for 15 minutes then remove them from the solution and dry them at 50 °C for 30 minutes. Take the a strip and

Nutrient broth to support the growth of bacteria

Gelatin as solidifying temperature and carbon source for bacteria,
solidifying

temperature is below 30°C.

pick bacteria by it. If the filter paper changed to violet this indicates that the bacteria is
oxidase positive, others is oxidase negative,

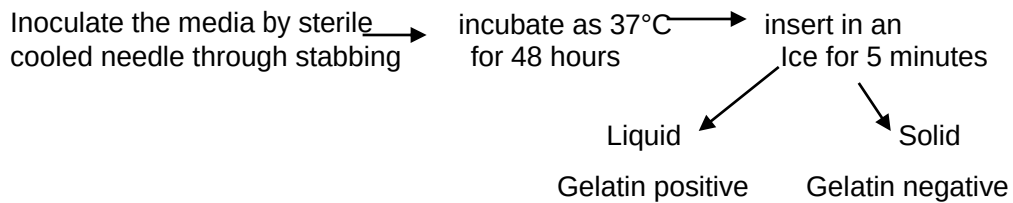
Extracellular enzymes

Gelatin test:

There are two methods:

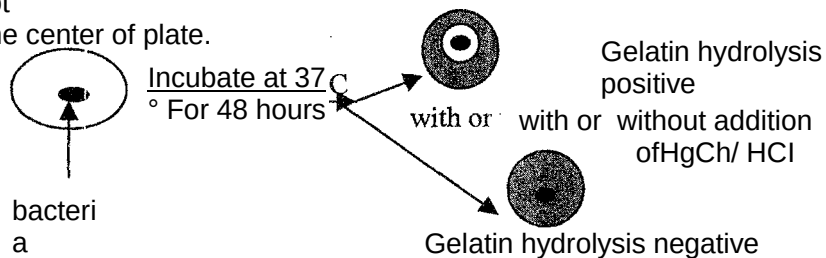
- 1- liquefaction
- 2- Plate methods

1- Liquefaction: the monomer of gelatin is amino acids



2- Plate methods:

Inoculate the nutrient agar + 4% gelatin with sterilized cooled loop from your isolate as a dot at the center of plate.



StarchTest: The monomer of starch is glucose

Inoculate the nutrient agar + 1 % starch with sterilized cooled loop from your isolate as a dot at the center of plate.

